

OPERATION AND MAINTENANCE MANUAL – PART 5

EXCITATION SYSTEM – CURATIVE MAINTENANCE

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Section 5 CURATIVE MAINTENANCE

5.1 General

The following section describes the troubleshooting system, and provides fault location procedures, starting with fault messages generated by the regulator. It helps maintenance operators locate system faults, from indications provided by the diagnostic system, and gives instructions for consequent repair.

To assess the correct operation or possible fault conditions of the equipment, diagnostic information is provided by:

Diodes on the regulator boards (inside the excitation cubicle),

Alarms displayed on the local control panel (LCP) located on the front door of the cubicle (see the relevant drawing),

Alarms displayed by the remote control system.

Both the remote and the local alarms can be used for fault location, but local alarms provide more useful details for troubleshooting.

Before you proceed with acknowledge of the defaults, the source and reason of the fault shall be clearly identified. Make sure that the fault is not caused by external equipment, for example by network disturbances. Then you can acknowledge the fault message (and eventually to reset the CPU board) from the Local Control Panel (LCP) and restart the excitation system. If the fault has disappeared, the system will immediately resume its normal operation. If the fault is still present, another alarm will be displayed.

Before replacing a faulty component (and particularly an electronic board), it is necessary to check the configuration and settings of the new component.

5.2 Safety requirements

Before any action detailed in this manual is carried out, the Site Health and Safety Instructions should be read and fully understood. The instructions in this document should also be read and fully understood.

Safety procedures have to be followed to lock the excitation system during operations inside the excitation cubicle by using a key switch or other equivalent devices.

High voltages and heavy currents are present inside the cubicle and on the excitation transformer. Accidental contact with live parts (field breaker, busbars,...) can cause serious injury.

Maintenance works should only be carried out with the excitation system switched off and all sensitive parts (field breaker, discharging resistors, ...) discharged. Before any work is carried out, the cubicle must be earthed.

Requirements to maintenance personnel

Whenever the equipment is operated, maintained or used in any way, the procedures in the Site Health and Safety Manual and any procedures detailed in these instructions should be followed.

All maintenance personnel must be trained on preventive and corrective maintenance.

They must be well acquainted with the excitation system, its power supply structure (redundancy, etc...), as well as its operating modes (local, remote ...).

Personnel must be informed and trained in emergency actions like emergency shut down, fire fighting etc...

Only authorised personnel may perform manipulations inside the excitation cubicle.

CAUTION: Electronic boards may be sensitive to electrostatic discharges. Maintenance technicians should wear anti-static equipment or they have to discharge themselves to an unpainted surface of the cubicle. Electronic boards must be stored in appropriate bags. Use an antistatic bench, connected to the ground of the cubicle, to work on the electronic boards.

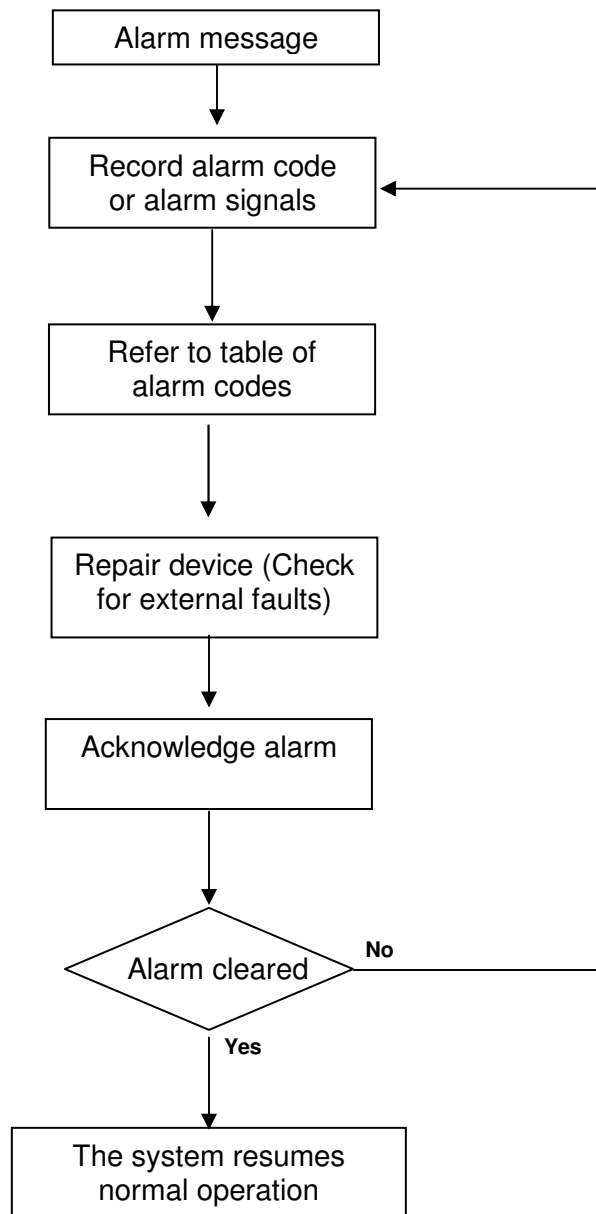
CAUTION: Before replacing a board, it is recommended to check that all boards are properly connected.

CAUTION: The hardware must be switched off before removing or installing a board.

5.3 Diagnostic

Fault detection is mainly based on alarm LEDs of the regulator boards, and on trouble shooting messages displayed on the Local Control Panel (LCP) or in the control room.

5.3.1 Fault diagnostic



Local fault finding facilities

Fault finding facilities are provided by LEDs in front of the different regulator boards, hence the information is accessible after opening the door of the regulator compartment in the excitation cubicle.

To avoid loss of useful troubleshooting information, you should try to analyse the faults before disconnecting the power supply from the digital voltage regulator.

The internal status of the electronic boards is also indicated by LEDs. Such LEDs are located on the regulator and regulation dedicated cards (Thyristor Triggering Module). The information is also accessible after opening the door of the regulator compartment in the excitation cubicle.

The main faults and status data are available on hard wired terminal board or on E8000 bus (for indication in the control room).

In addition to these indications, error messages are recorded and sent to the local control panel (LCP) and/or to the remote control system.

5.4 Internal led indicators

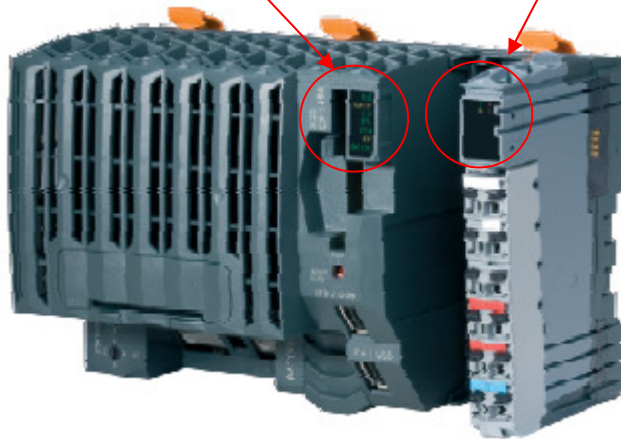
5.4.1 Module CP1485

The CPU module has got an integrated power supply module. Thus status leds are available at CPU and power supply location.

CPU Status leds



Power supply Status leds



CPU Status leds:

The different meaning of the leds is as follows:

	LED	Color	Status	Description
	R/E	Green	On	Application running
		Red	On	CPU initialising or system failure code (see hereafter)
	RDY/F	Yellow	On	CPU active
		Red	On	Overtemperature
	S/E	Green	On	Ethernet TCP/IP interface is configured on CPU.
		Red	On or blinking	Powerlink interface error
	EPL	Green	On	A link to the POWERLINK remote station has been established.
			Blinking	A link to the POWERLINK remote station has been established. The LED blinks when Ethernet activity is present on the bus.
	ETH	Green	On	A link to the Ethernet remote station has been established.
			Blinking	A link to the Ethernet remote station has been established. The LED blinks when Ethernet activity is present on the bus.
	CF	Yellow	On	CompactFlash OK
		Green	On	CompactFlash active
	DC OK	Green	On	CPU power supply OK
		Red	On	Backup battery is empty

System failure error codes

Incorrect configuration or defective hardware can cause a system failure error.

The error is displayed via the red error LED (R/E) using four switch-on phases. The switch-on phases are either 150 ms or 600 ms long. Error code outputs are repeated cyclically after 2 seconds.

Legend:	•	150 ms
	–	600 ms
	Pause	2 s delay

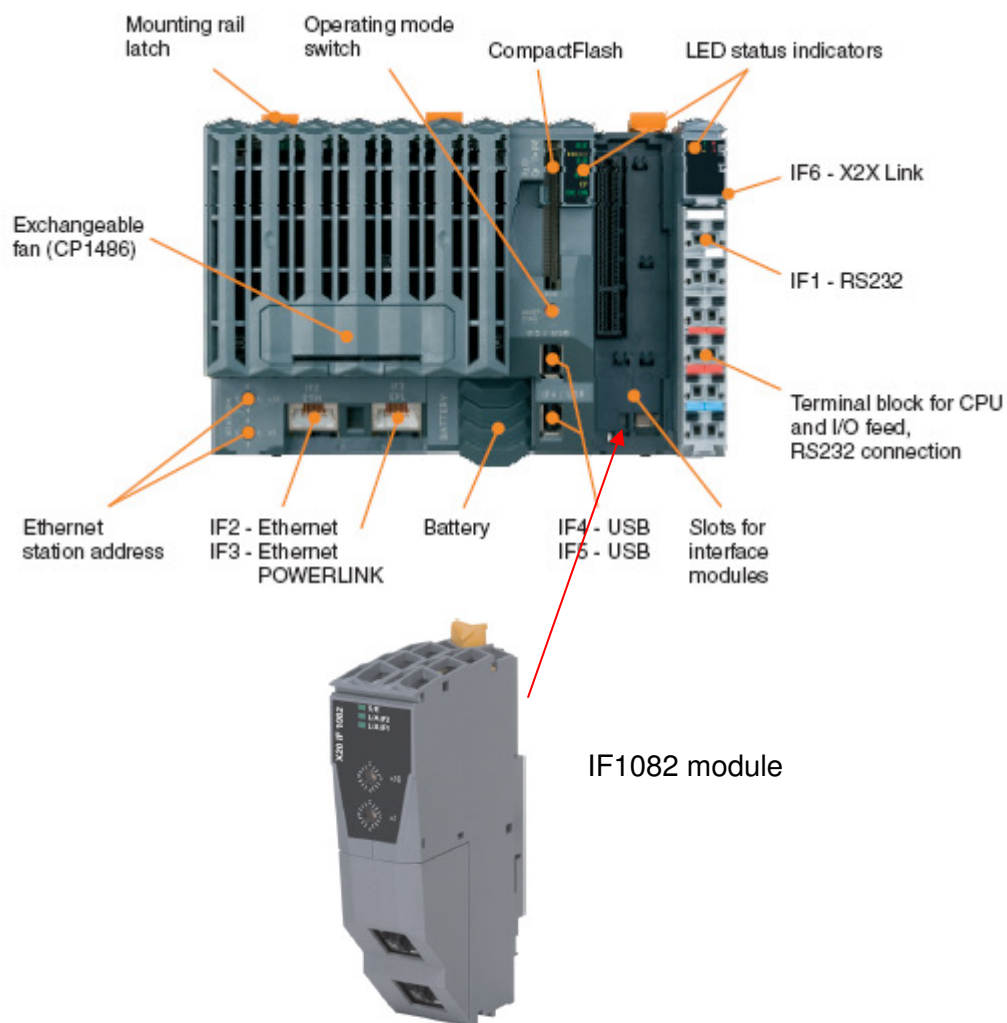
Error description	Error code displayed by red status LED							
RAM Error	•	•	•	–	Pause	•	•	•
Hardware Errors	–	•	•	–	Pause	–	•	•

Power Status leds:

	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: - X2X link power supply is overloaded - I/O supply is too low - Input voltage for X2X Link supply is too low
	e+r	Steady red / single green flash		Invalid firmware
	S	Yellow	Off	No RS232 activity
			On	The LED is on when data is being sent or received via the RS232 interface
	I	Red	Off	X2X link supply in the acceptable range
			On	X2X link power supply is overloaded

5.4.2 Module IF1082

A communication module is fitted within the CPU Interface module slot:



LED indicators provide the status of this communication module:

	LED	Color	Status	Description
	S/E	Green	On	Module is configured as TCP-IP interface
		Red	On	System failure (refer hereafter)
	L/A IFx	Green	On	A link to the remote station has been established.
			Blinking	A link to the remote station has been established. The LED blinks when Ethernet activity is present on the bus.

System failure error codes

Incorrect configuration or defective hardware can cause a system failure error.

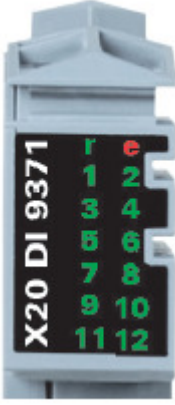
The error is displayed via the red error LED using four switch-on phases. The switch-on phases are either 150 ms or 600 ms long. Error code outputs are repeated cyclically after 2 seconds.

Legend:	•	150 ms
	–	600 ms
	Pause	2 s delay

Error description	Error code displayed by red status LED							
RAM Error	•	•	•	–	Pause	•	•	•
Hardware Errors	–	•	•	–	Pause	–	•	•

5.4.3 Module X20DI9371

Digital input module:

	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
	e+r	Steady red/ Single green flash		Invalid firmware
1-12		Green		Input status of the corresponding digital input

5.4.4 Module X20DO8332

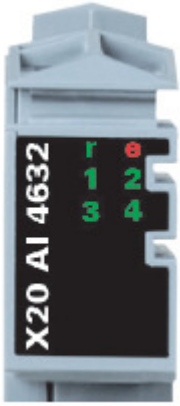
Digital output module:

Digital output module:

LED	Color	Status	Description
r	Green	Off	Module supply not connected
		Single flash	Reset mode
		Blinking	Preoperational mode
		On	RUN mode
e	Red	Off	Module supply not connected or everything is OK
		Single flash	Warning /error for an I/O channel. Level monitoring for digitals outputs has been triggered
		Double flash	I/O supply too low
e+r	Steady red/ Single green flash		Invalid firmware
1-8	Orange		Output status of the corresponding digital output

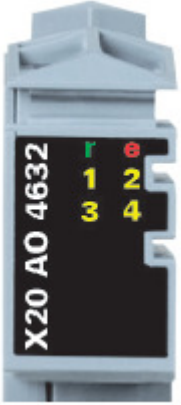
5.4.5 Module X20AI4632

Analog input module:

	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
			Double flash	System error: • Violation of the scan time • Synchronization error
	1-4	Green	Off	Open connection or sensor is disconnected
			On	The analog/digital converter is running, value is OK

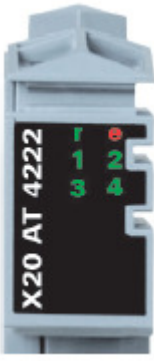
5.4.6 Module X20AO4632

Analog output module:

	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
	1-4	Orange	Off	Value=0
			On	Value ≠0

5.4.7 Module X20AT4222

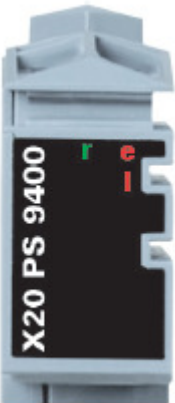
Module for PT100 temperature measurement.

	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs
	e+r	Steady red / single green flash		Invalid firmware
	1-4	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running

5.4.8 Module X20PS9400

Power supply module:

This module is dedicated to distributed IOS.

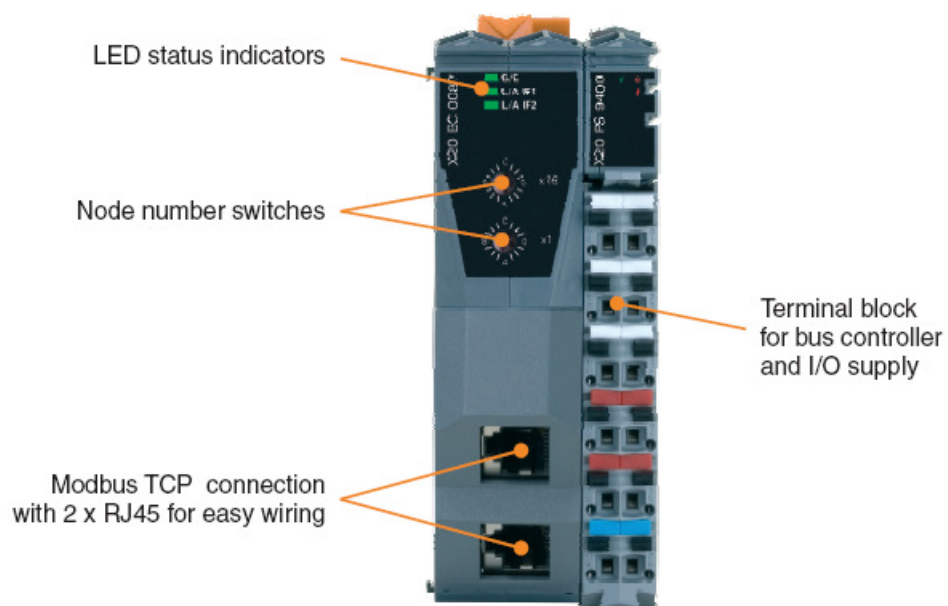
	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: - X2X link power supply is overloaded - I/O supply is too low - Input voltage for X2X Link supply is too low
	e+r	Steady red / single green flash		Invalid firmware
	l	Red	Off	X2X link supply in the acceptable range
			On	X2X link power supply is overloaded

5.4.9 Module X20BC0087

Distributed IO communication module :

	LED	Color	Status	Description
	S/E	Green	On	There is at least one client connection.
			2 pulses	No client connection exists.
			4 pulses	The controller is waiting for an address from the DHCP server.
			Blinking	Initialization of the connected I/O modules.
		Red	2 pulses	Watchdog timeout.
			4 pulses	The controller detected an IP address used twice.
			6 pulses	Error reading or writing flash memory.
			On	Irrecoverable hardware error.
	L/A IFx	Green	Blinking	The LED blinks when Ethernet activity is present on the bus.
			On	There is a connection (link), however no communication is taking place.
			Off	No physical Ethernet connection exists.

This module is associated with X20PS9400 module:



5.4.10 Module TTM211

Serial Ports:

Three serial ports are available on TTM211 module:

- P1 and P2 : RS232 Modbus serial link (RJ12).
- P3 : RS485 Serial link (DB15)

The serial port P1 is used to connect to the control panel with a dedicated program to change TTM211 internal settings and display internal faults of the module with a Modbus connection.

The Local Control Panel displays main modules faults and detailed faults are available on Maintenance Computer.

LEDs operation

This is the representation adopted in the tables to determine the LED status:

 : LED unlit.
  : LED flashing.
  : LED lit steadily.

LED « ON » at power up:

This LED denotes the operation of the module. It is green and marked « O » on the front panel of the module.

Green LED O	Meaning
	The green LED stays unlit. The TTM211 module is off or defective. ➔ Turn the power supply off then on again.
	The green LED is flashing. The TTM211 module is in the standby mode, or in the manual mode or in the local mode. It is energised and on standby.
	The green LED is steadily lighted. The module is energised. It is not in the standby mode.

LED Fault « D »

This LED indicates the module alarm status. It is red and identified « D » on the front of the module.

Red LED D	Meaning
	The red LED remains unlit. There is no current alarm on the TTM211 module.
	The red LED is flashing. There is a functional alarm on the TTM211 module. ➔ Check with the modbus connection to the module. An operational alarm is present on the module.
	The red LED is lit steadily. The module is in the module alarm state. ➔ Check t with the modbus connection to the module.. An operational or a hardware alarm is present on the module.

LED Operating mode « M »

This LED indicates that the TTM211 module is operating. It is orange and marked « M » on the front panel of the module.

Orange LED M	Meaning
	The orange LED stays unlit. The TTM211 module is in regulation mode.
	The orange LED flashes. The module is in field flashing mode or manual mode or test mode.
	The orange LED is steadily lit. The module is in local mode or manual mode.

Different case of LED status.

The following table represents several typical configurations of the LED status depending on the operational mode.

Green LED (ON)	Orange LED (MOD)	Red LED (DEF)	Signification
			Green LED steadily lit, orange and red LEDs unlit. Good operation, 'Auto AVR' mode.
			Green LED flashing, orange and red LEDs unlit. Good operation, 'stand-by' mode.
			Orange and green LEDs flash simultaneously, red LED unlit. 'Manual' mode.
		No meaning	Green LED flashing, orange LED steadily lit, red LED without meaning. 'Local' mode.
		No meaning	Green LED steadily lit, orange LED flashing, red LED without meaning. 'Flashing' mode.
			Orange LED unlit, green and red LEDs steadily lit. Blocking alarm in progress, 'Fallback' mode.
No meaning	No meaning		Green and orange LEDs without signification, red LED flashing. No-blocking alarm in progress.
			Green and red LEDs flash simultaneously, orange on opposition. 'Test' mode in progress.
			All three LEDs unlit. Module switched off or out of order. Apply voltage and if the problem persists replace the module.
			All three LEDs steadily lit during 1 s means that the module operates in self test, 'init' mode. If they permanent lit, the module waits in mode init for the disappearing of one alarm. If the serial link is unavailable the module is certainly out of order. Switch the power off and the apply voltage. If the problem persists replace the module.

Legend :  : LED unlit.  : LED flashing.  : LED lit steadily.

Modbus alarms display

Operational alarms:

Alarms on analogue setpoint line:

The analogue current setpoint is the 4-20mA type. The limit value range is 2-22mA. The following explains how the module behaves at the extreme values of the setpoint:

- Less than 2 mA causes regulation on a previous setpoint (i.e. 20 ms) for 100 ms (or less if it crosses the 4 mA threshold again). After 100ms, the setpoint moves to an equivalent theoretical value of 2 mA in % and a functional alarm (CRL: line break) is generated. The TTM211 module then changes to fallback mode when the parameter CxMa is equal to 0, otherwise it changes to manual mode or remain in stand-by mode.
- More than 22 mA causes regulation on a previous setpoint (i.e. 20 ms) for 100 ms (or less if it crosses the 20 mA threshold again). After 100ms, the setpoint moves to an equivalent theoretical value of 22 mA in % and a functional alarm (CSAT: saturated setpoint) is generated. The TTM211 module then changes to fallback mode when the parameter CxMa is equal to 0, otherwise it changes to manual mode or remain in stand-by mode.

« CRL Rupt. Line ana. setpoint » :

This variable informs the user of whether the analogue setpoint line has been broken (less than 2mA).

Indications given: . 0: FALSE (the analogue setpoint line is not broken).
. 1: TRUE (break in analogue setpoint line).

« CSAT Saturated ana. setpoint » :

This variable informs the user whether analogue setpoint line is saturated (more than 22mA).

Indications given: . 0: FALSE (the analogue setpoint line is not saturated).
. 1: TRUE (the analogue setpoint line is saturated).

Alarms concerning the acquisition of synchronisation signals:

The TTM211 module is designed to operate with rated frequency coil supply signals configured by the parameter FrN which lies between 40 Hz and 450 Hz.

It carries out passage detection through to zero of the power supply phases in order to directly control the thyristors for the coil current regulation. This also enables it to check the phase order.

The measurements made are tested by the TTM211 module, which makes it possible to generate the operational alarms in the event of a fault.

The Pmin, DPIT, DREF and DOPH variables indicate to the user the actual operational alarms and correspond to the following faults:

Invalid measured period range (Pmin).

Instantaneous measured period out of average range (DPIT).

One phase's amplitude is very different from the other (DREF).

Phase order incorrect on start-up or loss of phases when in service (DOPH).

The variables Pmin, DPIT and DOPH are evaluated only starting from one certain threshold, that is to say the measure MREF greater than the parameter SRFP.

« **PMin. Phases period min** »:

This variable informs the user whether the period of the coil power signal, which drives the thyristor bridge, is less than the limit value of 1.72 ms.

Indications given:

- . 0: FALSE → Signal period greater than the limit value of 1.72 ms. No alarm.
- . 1: TRUE → Signal period less than the limit value of 1.72 ms. An operational alarm is generated.

« **DPIT Phases period fault** »:

This variable informs the user if time between two interrupts leaves a given range value of the average measured period. This can happen if one phase signal is not symmetrical against the 0 volt or if the phase displacement between the signals is clearly different from 120 °.

This module cannot switch to the auto-AVR mode if this failure is detected on start up (fallback).

Indications given:

- . 0: FALSE → The 3 phase's signals are correct. No alarm.
- . 1: TRUE → One or more of the phase's signals are incorrect. An operational alarm is generated and the fallback mode is entered.

« **DREF Phases amplitude fault.** »:

This variable indicates to the user whether one phase signal presents one amplitude different from 20% against the others. This failure is only possible with a full wave three phase bridge.

This module cannot switch to the auto-AVR mode if this failure is detected on start up (fallback).

Indications given:

- . 0: FALSE → The 3 phase's signals are correct. No alarm.
- . 1: TRUE → One of the phase's signal amplitude is incorrect. A n operational alarm is generated and the fallback mode is entered.

« **DOPH Phases order fault** »:

This variable indicates to the user whether the synchronisation signals (phases of current power supply driving the bridge) are in right order. If it occurs when in operation progress, it signals a problem on the synchronisation signals, such as a phase loss or a short-circuit.

This module cannot switch to the auto-AVR mode if the phase order detected on start up is incorrect (fallback).

Indications given:

. 0	: FALSE	→	Phases in order.	No alarm.
. 1	: TRUE	→	Phases in disorder. An operational alarm is generated and the fallback mode is entered.	

Note:

When the parameter CxMa equals 1, the operational alarms CRL and CSAT and Pmin leads to flashing of the red LED « **D** ».

When the parameter CxMa equals 0, the operational alarms CRL and CSAT and all others (except Pmin) leads to light up steadily the red LED « **D** ».

Hardware alarms:

Any module alarm in progress will steadily light the LED D of the TTM211 module and set the alarm bits. However, the module alarm that indicates an internal malfunction in the module may not appear in some cases. In fact, the malfunctions in question may affect the serial communication with the supervisor or the displays on the LED's, making it impossible to diagnose the problem on these indicators.

« ItP1 It Parasite Level 1 » to « It Parasite Level 7 »:

These 7 alarm bits indicate unwanted interruptions at various levels. This type of fault appears when the interruption cycle, set at the indicated level, hasn't operated correctly (eg the initiating pulse of the interrupt routine is too short).

« Wd Watchdog (comm HS) »:

This fault appears when the watchdog drops out. The dropping out of the watchdog indicates the failure of the TTM211 module software. When this type of fault appears, communication with the module is interrupted.

« ErLo Software error »:

This alarm indicates a software error in the TTM211 module (failure) for an unknown task.

« IntA Interface absent »:

The TTM211 module is made of the MACO board and one interface board specific to the TTM211 module application (IP166 board). If the interface is absent or defective this alarm bit is set to 1.

« Intl Interface unknown »:

The MACO that contains the TTM211 module software is not assembled with the IP166 interface board.

« ECI Configuration incorrect »:

This fault appears when the EEPROM header on the MACO board is not compatible with the module in which it is installed.

This type of fault can result from an incorrect configuration (equipment identification bits) from the EEPROMs of the MACO board, or from their malfunctioning (EEPROMs out of order)

« **ReMo Reload of configuration** »:

This alarm indicates the reconfiguration of the TTM211 module by the MODBUS link is in progress. It corresponds to the downloading of data from the PC to the module. Wait for the end of reconfiguration before turning off or restarting the module.

« **VCI Version incorrect** »:

The version of the configuration installed on the TTM211 module is incorrect (old version).

Detection of memory fault:

The EEPROMs are designed to initialise the configuration in the RAM (Random access memory) on module start-up. The TTM211 module compares the working copy and the configuration contained in the EEPROMs to detect any latent defect (due to a simulated change of bytes). This may concern the failure of one of the two EEPROM memory banks, taking into account that one manages the even addresses and the other the odd ones.

« **CRPP Conf RAM != PROM Even** »:

The comparison of the configuration stored in the EEPROM with the working copy of the configuration in the RAM, reveals a possible latent defect in the configuration in the EEPROM at an even memory address.

« **CRPI Conf RAM != PROM Odd** »:

The comparison of the configuration stored in the EEPROM with the working copy of the configuration in the RAM, reveals a possible latent defect in the configuration in the EEPROM at an odd memory address.

« **MAJF Problems of flash update** »:

This fault indicates that it is impossible to update the flash memory (writing problem).

Alarm on acquisition channels:

The module contains internal analogue test channels (test 5%, 80% and offset) driving the ADC (Analogue Digital Converter). The corresponding digital outputs, also used for calibration, have to be contained within a certain range, as defined in the TTM211 module software in PtCAD. Outside this range, it can be considered that the acquisition chain in question (ADC) is defective. A corresponding alarm module is then generated.

« VT5 5% test channel fault»:

This alarm signals a fault on the 5% test channel. It denotes an analogue – digital conversion problem for the low analogue levels (bottom of range).

Expected ADC output value: 204 ± 60 PtCAD with hysteresis of 10 PtCAD.

« VT80 80% test channel fault»:

This alarm signals a fault on the 80% test channel. It denotes an analogue – digital conversion problem for the high analogue levels (top of range).

Expected ADC output value: 3276 ± 60 PtCAD with hysteresis of 10 PtCAD.

Detection of invalid memory configuration menu:

The TTM211 module has detected an error in one of the Parameter menus. The CRC check verifies the validity of the data received by each menu and therefore this signals the alarm in the event of an error (for instance – a problem during the writing of EEPROM). The alarm bits, which signal this type of fault are as follows.

« Men1 Menu 1 config. Invalid »:

Configuration menu 1 (GENERAL CONFIG) is not valid.

« Men2 Menu 2 config. Invalid »:

Configuration menu 2 (CURRENT CONTROL) is not valid.

« Men3 Menu 3 config. Invalid »:

Configuration menu 3 (SCALING PARAMETERS) is not valid.

5.4.11 HUB 8TX-ZF

	LED	Color	Description
	PW	Green	Supply voltage OK
	COL	Yellow	Telegram collision
	Port x	Green	Link active
		Yellow	Receiving telegrams

Local Configuration Options

At the "10M - 100M" slide switch at the front you have the option to set the data transmission rate for the entire device and thus for all connected devices to the fixed value of 10 Mbps or 100 Mbps.

At the "CROSS - STRAIGHT" slide switch at the front you have the option to set the port assignment for port 5 between 1:1 (STRAIGHT) and cross-over (CROSS). For this reason, port 5 is particularly suitable as uplink port.

Switch "10M -100M" for data transmission rate

Switch Position	Data Transmission Rate
10M	10 Mbps (default)
100M	100 Mbps



Please note that the data transmission rate that is selected using the "10M - 100M" switch is only valid after a reset (power-up) of the device.

"STRAIGHT - CROSS" switch for cable type

Switch Position	Cable Type
STRAIGHT	Normal (default)
CROSS	Uplink

5.4.12 FL SWITCH LM 8TX

	LED	Color	Status	Description
	US1	Green	On	Supply voltage US1 OK
	US2	Green	On	Supply voltage US2 OK
	LNK	Green	On	Link active
			Flashing	Transmitting/receiving
	100	Yellow	On	Full duplex mode
			Off	Half duplex mode
			Yellow	Collision detected

5.4.13 Local control panel

The table below gives the meanings of the Local Control Panel:

LED		Status	Description
Name	Colour		
PWR	RED		
		FLASHING	Unit is in the boot loader, no valid configuration is loaded. After downloading a configuration, if the light remains in the flashing state continuously, try cycling power. If the LED still continues to flash, try downloading a configuration again.
		STEADY	Unit is powered and running an application.
Middle	Yellow	OFF	No CompactFlash card is present.
		STEADY	Valid CompactFlash card present.
		FLASHING RAPIDLY	CompactFlash card being checked.
		FLICKERING	Unit is writing to the CompactFlash, either because it is storing data, or because the PC connected via the USB port has locked the drive. Do not turn off power to the unit while this light is flickering. The unit writes data in two minute intervals.
		FLASHING SLOWLY	Incorrectly formatted CompactFlash card present.
Bottom	Green	FLASHING	A tag is in an alarm state.
		STEADY	Valid configuration is loaded and there are no alarms present.
		LIT	LCP operating

5.5 Application and System faults

The faults are divided into first stage faults and second stage faults. First stage faults will produce an alarm and, for some of them, a changeover from the active regulator to the standby regulator. Second stage faults cause the field breaker to trip. Although the first stage faults do not lead to tripping of the unit, the problem should be investigated as soon as possible in order to keep the excitation system in good operation order.

The alarms are indicated on the local control panel (LCP) in front of the excitation cubicle.

Description of possible reasons hereafter is not exhaustive but constitutes a guide for maintenance.

All possible alarm and fault messages displayed by the Local Control Panel (LCP) are listed in the tables below. These alarms are available according to the project configuration.

Each alarm is displayed with a number, identifying completely the alarm.

From 1 to 299: Regulator's alarm.

From 300 to 599: Detailed rectifier's alarm.

From 600 to 699: Pulse generator's alarm.

From 1000 to 1200 Current sharing system (TPC) alarm

At the end of the text following numbers are displayed identifying the detection of fault:

"(1)": the alarm was detected by the primary regulator.

"(2)": the alarm was detected by the secondary regulator.

5.5.1 Regulator alarms

The following section describes all the alarms detected by the regulator.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
1: Maximum Stator Voltage Set Point	The voltage setpoint has reached its maximum value. Detection: Regulation software detection. Consequence: Voltage setpoint value is limited to its maximum value.	- Decrease the voltage setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
2: Minimum Stator Voltage Set Point	The voltage setpoint has reached its minimum value. Detection: Regulation software detection. Consequence: Voltage setpoint value is limited to its minimum value.	- Increase the voltage setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
3: Maximum Field Current Set Point	The current setpoint has reached its maximum value. Detection: Regulation software detection. Consequence: Current setpoint value is limited to its maximum value.	- Decrease the current setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
4: Minimum Field Current Set Point	The current setpoint has reached its minimum value. Detection: Regulation software detection. Consequence: Current setpoint value is limited to its minimum value.	- Increase the current setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
5: 1st Stage Field Over-Current	An excitation overcurrent has been detected by the overcurrent protection relay in the regulation compartment or by an internal fixed limit. Detection: Overcurrent protection relay or regulation software detection. Consequence: - Regulator change over (if possible). - Auto to manual mode change over (if possible).	• Control system fault. – Check the regulator output. – Check the TTM pulse generator board. • Sensor or wiring fault. – Check the wiring and the current sensor. – Check the sensor settings for compliance with the project specific factory test specification. CAUTION: A 2nd stage fault will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
6: 2nd Stage Field Over-Current	A first stage excitation overcurrent has been detected for a too long time, while still in manual mode. Detection: Regulation software detection. Consequence: - Trip demand of the unit. - Inverter forcing and field breaker opening after the unit breaker is opened.	• Check the circuit connected to the secondary of the excitation transformer and the excitation transformer. • Control system fault. – Check the regulator output. – Check the TTM pulse generator board. • Sensor or wiring fault. – Check the wiring and the current sensor. – Check the sensor settings according to the project specific factory test specification.
7: Field Over-Current in Manual Mode	A first stage excitation overcurrent has been detected for a too long time, while still in manual mode. Detection: Regulation software detection. Consequence: Local control panel alarm indication.	• Control system fault. – Check the regulator output. – Check the TTM pulse generator board. • Sensor or wiring fault. – Check the wiring and the current sensor. – Check the sensor settings according to the project specific factory test specification.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
8: Under-Excit. threshold exceeded	The under excitation threshold was exceeded: generator has been operating out of P-Q diagram. Detection: Regulation software detection. Consequence: - Regulator change over (if possible). - Auto to manual mode change over (if possible).	• The secondary winding of the excitation transformer is faulty. – Check the excitation transformer when the system is stopped. • Control system fault. – Check the regulator output. – Check the TTM pulse generator board. – Check the stator voltage and stator current measurement modules. • Stator measurement fault – Check the measurement of stator current and voltage.
9: Maximum Tan PHI Set Point	The Tan PHI setpoint has reached its maximum value. Detection: Consequence: Tan PHI setpoint value is limited to its maximum value.	- Decrease the Tan PHI setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
10: Minimum Tan PHI Set Point	The Tan PHI setpoint has reached its minimum value. Detection: Regulation software detection. Consequence: Tan PHI setpoint value is limited to its minimum value.	- Increase the Tan PHI setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
11: Maximum Q Set Point	The reactive power setpoint has reached its maximum value. Detection: Regulation software detection. Consequence: Reactive power setpoint value is limited to its maximum value.	- Decrease the reactive power setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
12: Minimum Q Set Point	The reactive power setpoint has reached its minimum value. Detection: Regulation software detection. Consequence: Reactive power setpoint value is limited to its minimum value.	- Increase the reactive power setpoint value (with Local Control Panel or remote control system). - Check the parameters settings for compliance with the project specification.
13: U/F Maximum Limitation Active	The U/F value has reached its maximum value. Detection: Regulation software detection. Consequence: Voltage setpoint value is limited as per U/F maximum value.	- Decrease the voltage setpoint or increase the frequency of the unit. - Check the parameters settings for compliance with the project specification.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
14: U/F Minimum Limitation Active	The U/F value has reached its minimum value. Detection: Regulation software detection. Consequence: Voltage setpoint value is limited as per U/F minimum value.	- Increase the voltage setpoint or decrease the frequency of the unit. - Check the parameters settings for compliance with the project specification.
15: Stator Current Limitation Active	The stator current has reached its maximum value and is limited by the regulator. Detection: Regulation software detection. Consequence: Limitation of stator current to its maximum value.	- Increase the stator voltage setpoint, change reactive power setpoint towards 0 or change Tan phi setpoint towards 0 as per regulation mode. - Decrease the active power of the unit. - Check the parameters settings for compliance with the project specification. CAUTION: A reduction of reactive power to 0 or Tan phi to 0 will disable the stator current limitation. Decrease of the active power can be necessary.
16: Over-Excit. Limitation Active	The rotor current has reached its maximum value and is limited by the regulator. Detection: Regulation software detection. Consequence: Limitation of rotor current to its maximum value.	- Decrease the stator voltage setpoint, decrease reactive power setpoint or decrease Tan phi setpoint as per regulation mode. - Check the parameters settings for compliance with the project specification.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
17: Under-Excit. Limitation Active	The rotor current has reached its minimum value and is limited by the regulator. Detection: Regulation software detection. Consequence: Limitation of rotor current to its minimum value.	- Increase the stator voltage setpoint, increase reactive power setpoint or increase Tan phi setpoint as per regulation mode. - Check the parameters settings for compliance with the project specification.
18: Max Corrected Stator Voltage Spt	The result voltage setpoint after limitation has reached its maximum value. Detection: Regulation software detection. Consequence: Regulated stator voltage is limited to its maximum value to stay within machine operating range.	- Check the parameters settings for compliance with the project specification. - Change the machine operating point (P-Q).
19: Min Corrected Stator Voltage Spt	The result voltage setpoint after limitation has reached its minimum value. Detection: Regulation software detection. Consequence: Regulated stator voltage is limited to its minimum value to stay within machine operating range.	- Check the parameters settings for compliance with the project specification. - Change the machine operating point (P-Q).

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
20: Thyristor Bridge Nr 1 fault	A fault has been detected in thyristor bridge -U21. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U21 thyristor bridge in the excitation cubicle. – Refer to the specific fault type indicated on the Local Control Panel (LCP). • Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.
21: Thyristor Bridge Nr 2 fault	A fault has been detected in thyristor bridge -U22. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U22 thyristor bridge in the excitation cubicle. • Refer to the specific fault type indicated on the Local Control Panel (LCP). Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
22: Thyristor Bridge Nr 3 fault	A fault has been detected in thyristor bridge -U23. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U23 thyristor bridge in the excitation cubicle. – Refer to the specific fault type indicated on the Local Control Panel (LCP). • Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.
23: Thyristor Bridge Nr 4 fault	A fault has been detected in thyristor bridge -U24. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U24 thyristor bridge in the excitation cubicle. – Refer to the specific fault type indicated on the Local Control Panel (LCP). • Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
24: Thyristor Bridge Nr 5 fault	A fault has been detected in thyristor bridge -U25. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U25 thyristor bridge in the excitation cubicle. – Refer to the specific fault type indicated on the Local Control Panel (LCP). • Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.
25: Thyristor Bridge Nr 6 fault	A fault has been detected in thyristor bridge -U26. Detection: Sensors in bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s) (if possible).	• Faulty -U26 thyristor bridge in the excitation cubicle. – Refer to the specific fault type indicated on the Local Control Panel (LCP). • Fuse fault • Cooling fault • Temperature fault • Conduction fault. • Isolation of bridge. • Pulse amplifier fault. • Pulse fault. • Faulty circuit downstream from the thyristor. – Check the snubber circuit – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
26: Thyristor Bridge Fan Failure	A fault has been detected on the cooling fan of a thyristor bridge by fuses or thermal relays. Thermal protection threshold: refer to the Preventive maintenance section. Detection: Relays and fan sensors. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• Cooling fault in one thyristor bridge of the excitation cubicle. Refer to the Local Control Panel (LCP) to identify the faulty bridge. – Check the relevant cooling fan motor and the cooling command sequence in the excitation cubicle. – Check the wiring and the transformer for a short circuit. – Change the faulty fuses. – Check and reset the overload relay. CAUTION: A fault in another bridge will cause the unit to trip.
27: Thyristor Bridge Fuses	A fuse is blown in one thyristor bridge. Detection: Fuse blowing detector in one bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• Faulty fuse in one thyristor bridge of the excitation cubicle. Refer to the Local Control Panel (LCP) to identify the faulty bridge. – Check the thyristors of the bridge. Change the faulty thyristors and fuses. • Faulty circuit downstream from the thyristor. – Check the downstream circuit. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
28: Thyristor Bridge Over-temp.	A temperature fault has been detected on one thyristor bridge. Upper threshold: refer to the Preventive maintenance section. Detection: Thermostat on top of the bridge, or temperature switch on the heat sink. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• Temperature fault in a thyristor bridge of the excitation cubicle. Refer to the Local Control Panel (LCP) to identify the faulty bridge. – Check the cooling air inlet filters and the air outlet. • Temperature fault due to overloading of one phase. – Check the current sharing between the bridges. – Check all connections of the bridge (internal and external busbars). CAUTION: A fault in another bridge will cause the unit to trip.
29: Excit. Transformer Over-temp, Alarm	Overheating of excitation transformer (1st stage fault) Detection: Temperature sensor in the secondary winding of the excitation transformer. Consequence: Alarm.	• The winding temperature of the excitation transformer exceeds the first upper threshold defined by the corresponding temperature sensor. – Check the cooling air inlet and outlet grids. – Check the ambient temperature in the transformer building or the cooling system. • The temperature measuring circuit is faulty. – Check the measuring circuit for damaged or disconnected wires, when the system is stopped. CAUTION: A 2nd stage fault will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
30: Excit. Transformer Over-Temp, Trip	Overheating of excitation transformer (2nd stage fault) Detection: Temperature sensor in the secondary winding of the excitation transformer. Consequence: Unit tripping demand. Inverter forcing and field breaker opening after the unit breaker is opened.	• The winding temperature of the excitation transformer exceeds the second upper threshold defined by the corresponding temperature sensor. – Check the cooling air inlet and outlet grids. – Check the ambient temperature in the transformer building or the cooling system. • The temperature measuring circuit is faulty. – Check the measuring circuit for damaged or disconnected wires, when the system is stopped.
31: Rotor Over-Voltage - Crowbar	An overvoltage has been detected across the rotor field windings. Detection: CROB1 board on crowbar unit. Consequence: Inverter forcing and field breaker opening after the unit breaker is opened.	• The crowbar system which protects the rotor against overvoltage is faulty. – Check the crowbar system. • Transient operation of the generator (loss of synchronisation) or generator fault. – Refer to the Generator Operation and Maintenance manual.
32: Regulator fault	Displayed regulator is faulty while the other regulator is controlling the generator. Detection: Regulation software detection. Consequence: Alarm.	• LCP displayed regulator has detected a fault. – Refer to the alarm text displayed on the Local Control Panel (LCP).

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
33: Rotor Insulation fault, 1st Stage	A current is flowing through the test brush and the ground. Detection: Protection relay in the regulator compartment. Consequence: Resistance below threshold: alarm indication.	• Rotor winding insulation fault. – Check the rotor insulation when the system is stopped. (Refer to the Generator Operation and Maintenance manual). • Slip ring or brush-holder system fault. – Check and clean the slip ring and brush-holder assembly when the system is stopped. (Refer to the Generator Operation and Maintenance manual). • Measuring circuit fault. – Check wiring of the measuring circuit. CAUTION: Operating with a rotor winding earth fault is dangerous, since a second fault would result in a short circuit causing severe damage due to the high D.C. power. Reduce the generator power and stop the unit as soon as possible.
34: Rotor Insulation fault, 2nd Stage	An earth fault current flowing through test brush applied to the generator shaft energises the relay. Detection: Protection relay in the regulator compartment. Consequence: 2nd stage fault (very low resistance threshold). Trip demand. Inverter forcing and field breaker opening after the unit breaker is opened.	• Rotor winding insulation fault. – Check the rotor insulation when the system is stopped. (Refer to the Generator Operation and Maintenance manual). • Slip ring or brush-holder system fault. – Check and clean the slip ring and brush-holder assembly when the system is stopped. (Refer to the Generator Operation and Maintenance manual). • Measuring circuit fault. – Check wiring of the measuring circuit.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
35: Flashing Fuse Fault	The fuse located in the flashing circuit has been detected as faulty. Detection: Flashing fuse. Consequence: The field breaker cannot be closed. A trip order is given if the field breaker is opened.	• Flashing fuse was blown. – Check the fuse and the flashing circuit (insulation of the flashing circuit). – Replace the defective fuse
36: Flashing too long	The duration of the field flashing sequence exceeds the TMPAM1 timing parameter. Detection: Regulation software detection. Consequence: Inverter forcing and field breaker opening.	• Faulty flashing circuit (stator voltage still below voltage threshold when time delay is over). – Check closing of flashing breaker and field breaker. – Check presence of flashing voltage. – Check the generator voltage measuring circuit.
37: Synchronization Voltage Fuses	The fuse for synchronisation voltage to the TTM module has been detected as faulty. Detection: Synchronisation voltage fuse. Consequence: Regulator changeover (if possible).	• Synchronisation voltage fuse was blown. – Check the fuse and the synchronisation voltage circuit (insulation of the circuit). – Replace the defective fuse

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
38: Voltage Measure Fault - External	External stator voltage measurement is not valid. Detection: Microbreaker in voltage measuring circuit. Consequence: Regulator changeover (if possible). Regulator auto to manu regulation mode changeover.	• Short-circuit in the stator voltage measuring circuit. – Check the voltage at test point
39: Pulse generator fault	Pulses generator is faulty Detection: The fault is detected by the pulses generator TTM (or PWM) of the regulator. Logical output Consequence: Regulator change over (if possible).	• Displayed regulator pulses generator in the excitation cubicle is faulty. – Check Pulse generator 24 V power supply (LED ON lit or flashing). – Refer to the alarm text displayed on the Local Control Panel (LCP). – Look at the TTM (or PWM) LED's (refer to § 5.4). – Connect with the maintenance to the module for further information (Refer to § 5.4). CAUTION: A fault occurring in the standby regulator will cause the unit to trip.
40: Pulse generator, 24 V DC feeder fault	Power supply to the TTM (or PWM) module is faulty. Detection: Microbreaker on TTM (or PWM) module power supply. Consequence: Regulator changeover (if possible).	• Power supply of the TTM (or PWM) module – Check the power supply circuit of the TTM (or PWM) module CAUTION: A fault occurring in the standby regulator will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
41: Second Stage Excitation Fault	Second stage excitation fault detected. Detection: Relays. Consequence: Unit trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	• No regulator active. • No regulator watchdog – Refer to the alarm text displayed on the Local Control Panel (LCP).
42: 1st Stage Exc Over-Current ext detect	An excitation overcurrent has been detected by the overcurrent protection relay in the regulation compartment. Detection: Protection relay. Consequence: Regulator changeover (if possible). Regulator Auto mode to Manu mode changeover.	• Control system fault. – Check the regulator output. – Check the TTM pulse generator board. • Sensor or wiring fault. – Check the wiring and the current sensor. – Check the sensor settings for compliance with the project specific factory test specification. CAUTION: A 2nd stage fault will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
43: 2nd Stage Excitation Over-Current	An excitation overcurrent has been detected by the overcurrent protection relay in the regulation compartment. Detection: Protection relay. Consequence: Unit trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	• Check the circuit connected to the secondary of the excitation transformer and the excitation transformer. • Control system fault. – Check the regulator output. – Check the TTM pulse generator board. • Sensor or wiring fault. – Check the wiring and the current sensor. – Check the sensor settings according to the project specific factory test specification.
44: Clipper Fuse Fault	A fuse has been detected as faulty in the overvoltage protection device. Detection: Fuse. Consequence: Alarm.	• Short-circuited component in the overvoltage protection device. – Check the diode bridge and the RC circuit when the system is stopped. Note: Closing the field breaker will not be possible unless the default is repaired.
45: 24V Supply Fault	Faulty 24 V power supply . Detection: Micro circuit breaker on 24 V power supply. Consequence: Alarm.	• Power supply fault detected by micro circuit breaker. – Check the 24 V power supply circuit. Note: Closing the field breaker will not be possible unless the fault is repaired.
46: Excitation 1st stage fault	1st stage excitation fault. Detection: Relays. Consequence: Alarm.	- Check the corresponding relays.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
47: 2nd regulator fault	Standby regulator is faulty while active regulator is controlling the generator. Detection: Regulation software detection. Consequence: Standby regulator not available.	- Active regulator has detected a fault on standby regulator. - Check power supply of the second regulator (LED PWR of PSD040 is green). - Select the "STANDBY VISU" to see alarms displayed on standby regulator. - Refer to the alarm text displayed on the Local Control Panel (LCP).
48: 2nd regulator watch dog	Standby regulator is faulty. Detection: Regulation software detection. Consequence: Standby regulator not available.	- Standby Regulator in the excitation cubicle is faulty. - Check the power supply circuit of the regulator. - Refer to LED's display on regulator modules. - Refer to the alarm text displayed on the Local Control Panel (LCP). CAUTION: A fault occurring in standby regulator will cause the unit to trip.
49: Measurement device Feeder Fault	Power supply to the measurement device is faulty. Detection: Microbreaker on measurement device power supply. Consequence: Alarm display. Inhibition of measurement device fault alarm.	- Power supply of the measurement device - Check the power supply circuit of the measurement device. CAUTION: A fault occurring in the standby regulator will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
50: Channel 1, DC Feeder Fault	Faulty 125 VDC power supply on channel 1. Detection: Micro circuit breaker on 125 VDC power supply. Consequence: Alarm.	- Power supply fault detected by micro circuit breaker on channel 1. – Check the 125 VDC power supply circuit.
51: Channel 2, DC Feeder Fault	Faulty 125 VDC power supply on channel 2. Detection: Micro circuit breaker on 125 VDC power supply. Consequence: Alarm.	- Power supply fault detected by micro circuit breaker on channel 2. – Check the 125 VDC power supply circuit.
52: Channel 1, AC Feeder Fault	Faulty AC power supply on channel 1. Detection: Fuses on AC power supply. Consequence: Alarm.	- Power supply fault detected by fuses on channel 1. – Check the AC power supply circuit.
53: Channel 2, AC Feeder Fault	Faulty AC power supply on channel 2. Detection: fuses on AC power supply. Consequence: Alarm.	- Power supply fault detected by fuses breaker on channel 2. – Check the AC power supply circuit.
54: Battery 1, Supply Fault	Faulty battery power supply on channel 1. Detection: Voltage monitoring relay on battery power supply. Consequence: Alarm.	- Power supply fault detected by voltage monitoring relay on channel 1. – Check the battery power supply circuit.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
55: Battery 2, Supply Fault	Faulty battery power supply on channel 2. Detection: Micro circuit breaker on battery power supply. Consequence: Alarm.	• Power supply fault detected by voltage monitoring relay on channel 2. – Check the battery power supply circuit.
56: Protection, Supply Fault	Faulty protection power supply. Detection: Micro circuit breaker on protection power supply. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	• Power supply fault detected by micro circuit breaker. – Check the protection power supply circuit.
57: UST meas fault	Faulty stator voltage measurement. Detection: UMT162 module. Consequence: Regulator changeover (if possible). Auto mode to manual mode changeover.	• A difference of at least 30% between the amplitude of one of the three voltages and the other two was detected. – Check the voltage measurement circuit. – Refer to LED's display on regulator stator voltage measurement module.
58: IST meas fault	Faulty stator current measurement. Detection: UMT162 module. Consequence: Regulator changeover (if possible). Auto mode to manual mode changeover.	• A difference of at least 30% between the amplitude of one of the three currents and the other two was detected. – Check the current measurement circuit. – Refer to LED's display on regulator stator current measurement module.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
59: Rotor over-temp 2nd stage	2nd stage rotor overtemperature detection. Detection: Regulation software detection. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- The calculated rotor temperature value has increased above its 2nd stage threshold value - Refer to the Generator Operation and Maintenance manual. - Check the rotor current measurement circuit.
60: Rotor over-temp 1st stage	1st stage rotor overtemperature detection. Detection: Regulation software detection. Consequence: Alarm	- The calculated rotor temperature value has increased above its 1st stage threshold value - Decrease excitation current.
61: DC breaker discrepancy	Discrepancy in field breaker position. Detection: Field breaker position switches. Consequence: Alarm	- A discrepancy in field breaker position was detected. The field breaker position does not match the field breaker command or its position is detected neither opened nor closed: - Check contacts of the field breaker position (opened and closed contacts). - Check operation of the field breaker. Note: Closing the field breaker will not be possible unless the default is repaired.
62: Alarm	Alarm indication Detection: Regulation software detection. Consequence: Alarm indication.	- An alarm has been detected - Refer to the alarm text displayed on the Local Control Panel (LCP).

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
63: Trip	Trip request. Detection: Regulation software detection. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	• A trip is being requested due to excitation system fault. – Refer to the alarm text displayed on the Local Control Panel (LCP).
64: Power bridge supply fault	Faulty thyristor bridge power supply. Detection: External relay. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	• Power supply fault detected at the excitation transformer level. – Check the power supply of the excitation transformer.
65: Reg application fault	Displayed Regulator has got an application fault. Detection: Regulation software detection. Consequence: Regulator changeover (if possible).	• An internal fault in the regulator has been detected. – Check the parameters loaded in CPU. – At unit stopped, reload the regulator program. – If problem is still present, change the CPU.
66: Homolog reg application fault	Homolog regulator has got an application fault. Detection: Regulation software detection. Consequence: Standby regulator not available.	• An internal fault in the homolog regulator has been detected. – Check the parameters loaded in CPU of homolog regulator. – At unit stopped, reload the homolog regulator program. – If problem still present, change the CPU.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
67: Regulator change-over by UST	Change over due to UST measurement difference between the two regulators. Detection: Regulation software detection. Consequence: Automatic change over to regulator with the highest measurement.	• A change over due to UST measurement difference between the two regulators. Occurred: – Check the stator voltage measurement circuits. – Refer to the alarm texts displayed on the Local Control Panel (LCP).
68: Module fault	Displayed Regulator has got a module fault. Detection: Regulation software detection. Consequence: Regulator changeover (if possible).	• An regulator module fault has been detected. – Refer to LED's display on regulator modules. – Refer to the alarm text displayed on the Local Control Panel (LCP). – Check the IO fault table With P80
69: Homolog reg module fault	Homolog Regulator has got a module fault. Detection: Regulation software detection. Consequence: Alarm.	• An regulator module fault has been detected. – Refer to LED's display on regulator modules. – Refer to the alarm text displayed on the Local Control Panel (LCP). – Check the IO fault table With P80
70: 1st stage buchholz	1st stage buchholz in the excitation transformer. Detection: Excitation transformer protection device. Consequence: Alarm.	• The excitation transformer is faulty. – Check the excitation transformer when the system is stopped. CAUTION: A 2nd stage Buchholtz fault will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
71: Excit transf oil level inf min	Excitation transformer low oil level. Detection: Excitation transformer oil level measurement device. Consequence: Alarm.	- The excitation transformer oil level is too low. - Refer to excitation transformer maintenance guide.
72: 2nd stage buchholz	2nd stage buchholz in the excitation transformer. Detection: Excitation transformer protection device. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- The excitation transformer is faulty. - Check the excitation transformer when the system is stopped.
73: 1st stage U/F protection	1st stage U/F protection. Detection: U/F protection relay. Consequence: Alarm.	- The U/F 1st stage threshold has been exceeded. - Decrease the voltage setpoint or increase the frequency of the unit. CAUTION: A 2nd stage U/F fault will cause the unit to trip.
74: 2nd stage U/F protection	2nd stage U/F protection. Detection: U/F protection relay. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- The U/F 2nd stage threshold has been exceeded. - Decrease the voltage setpoint or increase the frequency of the unit.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
75: DC Control Supply Fault	Faulty DC control power supply . Detection: Micro circuit breaker on DC power supply. Consequence: Trip request. Inverter forcing.	- Power supply fault detected by micro circuit breaker. - Check the DC control power supply circuit.
76: Manu Ustat limitation active	The Stator voltage value has reached its maximum value. Detection: Regulation software detection. Consequence: Excitation current setpoint value is limited as per stator voltage maximum value.	- Decrease the rotor current setpoint. - Check the parameters settings for compliance with the project specification.
77: Protection Relay Watch Dog	Protection relay watchdog. Detection: Protection relay. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- Protection relay watchdog - Check the protection relay operation.
78: Flashing Inhibition	Flashing inhibition. Detection: Digital input of the regulator. Consequence: Flashing inhibition.	External command for flashing inhibition.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
79: Local Remote discrepancy	Local remote discrepancy. Detection: Local/remote button. Consequence: Alarm.	- Local control and remote control are selected at the same time. - Check the local/remote selector.
80: Rot earth prot fuse flt	Rotor earth protection fuse fault. Detection: Fuse. Consequence: Rotor earth fault protection inhibition.	- Fuse of rotor earth fault protection is blown. - Check the rotor earth fault injection circuit. - Check the rotor insulation when the system is stopped. (Refer to the Generator Operation and Maintenance manual).
81: AVR change over upon failure	Regulator change over occurred due to a fault on the other regulator. Detection: Regulation software detection. Consequence: Alarm.	- A regulator change over occurred due to a fault to the other regulator: - Select the standby regulator visu. - Display alarms (on the standby regulator) and refer to the alarm text displayed on the Local Control Panel (LCP).
82: MX3 UPS supply fault	MX3 protection supply fault. Detection: MX3 protection. Consequence: Alarm.	- Power supply fault detected by MX3 protection. - Check the MX3 power supply circuit.
83: Grid measurement Failure	Grid measurement is not valid. Detection: Regulation software detection. Consequence: Deselection of grid regulating values.	4-20 mA signal received for active and reactive power are out of range. - Check the 4-20mA signals of active and reactive power. - Check the regulator analogue input card LED's status.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
84: Rotor current measure fault	Rotor current measurement fault. Detection: Rotor current measurement. Consequence: Invalid rotor current measurement. Regulator change over (if possible). Rotor temperature calculation unavailable. Manual mode unavailable. Underexcitation limitation, unavailable.	• Rotor current measurement fault – Check the rotor current transducer. – Check the rotor current measurement circuit. – Check the rotor current measurement module • Power supply fault detected by IEX164 card (if used in the system). – Check the IEX164 power supply circuit. CAUTION: On manual mode, this fault will cause the unit to trip. An auto mode fault, forcing to manual mode will cause the unit to trip. Underexcitation limitation is no more available.
85: Thyristor 1 conduction flt	Non conduction of thyristor No. 1 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	• Thyristor No. 1 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. – Check wiring from the pulse generator circuit. – Check the gate circuit components. – Check the busbar connections of the bridge. – Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
86: Thyristor 2 conduction flt	Non conduction of thyristor No. 2 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	• Thyristor No. 2 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. – Check wiring from the pulse generator circuit. – Check the gate circuit components. – Check the busbar connections of the bridge. – Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
87: Thyristor 3 conduction flt	Non conduction of thyristor No. 3 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	• Thyristor No. 3 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. – Check wiring from the pulse generator circuit. – Check the gate circuit components. – Check the busbar connections of the bridge. – Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
88: Thyristor 4 conduction flt	Non conduction of thyristor No. 4 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	• Thyristor No. 4 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. – Check wiring from the pulse generator circuit. – Check the gate circuit components. – Check the busbar connections of the bridge. – Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
89: Thyristor 5 conduction flt	Non conduction of thyristor No. 5 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	• Thyristor No. 5 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. – Check wiring from the pulse generator circuit. – Check the gate circuit components. – Check the busbar connections of the bridge. – Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
90: Thyristor 6 conduction flt	Non conduction of thyristor No. 6 in a rectifier bridge Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Alarm.	- Thyristor No. 6 of a thyristor bridge in the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty rectifier bridge. - Check wiring from the pulse generator circuit. - Check the gate circuit components. - Check the busbar connections of the bridge. - Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
91: Power converter 1 conduction flt	Non conduction of thyristor in thyristor bridge 1. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	- A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. - Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. - Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
92: Power converter 2 conduction flt	Non conduction of thyristor in thyristor bridge 2. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	- A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. - Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. - Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
93: Power converter 3 conduction flt	Non conduction of thyristor in thyristor bridge 3. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. – Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. – Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
94: Power converter 4 conduction flt	Non conduction of thyristor in thyristor bridge 4. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. – Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. – Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
95: Power converter 5 conduction flt	Non conduction of thyristor in thyristor bridge 5. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	• A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. – Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. – Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
96: Power converter 6 conduction flt	Non conduction of thyristor in thyristor bridge 6. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge. Consequence: Inhibition of firing pulses to the thyristor bridge: operation with the other bridge(s).	- A thyristor in a rectifier bridge of the excitation cubicle is faulty. Refer to the Local Control Panel (LCP) to identify the faulty bridge and the faulty thyristor. - Check the thyristor firing circuit and the pulse generator circuit, especially the gate circuit components. - Check the busbar connections of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
97: SUCO controller watchdog	Watchdog fault of the conduction monitoring PLC Detection: CPU board of the conduction monitoring PLC. Consequence: Alarm.	- The conduction monitoring PLC in the excitation cubicle is faulty. - Check and if necessary replace the CPU board. - Check and if necessary replace the PLC power supply. CAUTION: Switch off the power supply to the conduction monitoring PLC if an electronic board needs to be changed.
98: Field breaker closing speed flt	Field breaker closing speed fault. Detection: Regulation software detection. Consequence: Alarm.	- The field breaker closing command was not taken into account because speed is not reached. - Check speed signal.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
99: Delta UC flt on changeover rqst	Delta UC fault on changeover request. Detection: Regulation software detection. Consequence: Alarm. Standby regulator not available.	- The changeover command was not taken into account because of a difference between the two UC command. - Check wiring of UC signal between the two regulators. - Check parameter setting of the regulators. - Refer to LED's display on regulator modules.
100: Start up current too low	Back to back starting up current too low. Detection: Back to back protection relay. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- The back to back starting up current is too low. - Refer to the Generator Operation and Maintenance manual before restarting the unit.
101: Start up current too high	Back to back starting up current too high. Detection: Back to back protection relay. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	- The back to back starting up current is too high. - Refer to the Generator Operation and Maintenance manual before restarting the unit.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
102: Change over: Modbus line fault	Change over on Modbus line fault. Detection: Regulation software detection. Consequence: Regulator changeover.	• The modbus line has been lost. – Check the modbus line settings. – Check the modbus line life variable. – Check the modbus line commands on both regulators. CAUTION: On second modbus line on other regulator a changeover command will be issued again.
103: Pulse amplifier Fault	Fault on a TPA213 module. Detection: MOK signal from the TPA213 module. Consequence: Alarm. Bridge blocking. Bridge changeover (TWIN or redundant TWIN configuration).	• A TPA213 module has been detected faulty. Refer to the Local Control Panel (LCP) to identify the TPA213 module. – Check the corresponding TPA213 power supplies. CAUTION: On another TPA213 module fault, the unit will trip.
104: Ventilation flt	Ventilation fault on a bridge. Detection: Pressure detector on bridge. Consequence: Alarm. Bridge blocking. Bridge changeover (TWIN or redundant TWIN configuration).	• Refer to the Local Control Panel (LCP) to identify the faulty bridge. – Check operation of the fan and rotating direction. – Check the corresponding pressure detector. CAUTION: On another ventilation fault, the unit will trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
105: Pulse flt	Pulses are missing on one bridge. Detection: TPA213 pulse fault output. Consequence: Alarm. Bridge blocking. Bridge changeover (TWIN or redundant TWIN configuration).	- A pulse fault has been detected by the TPA213 module. Refer to the Local Control Panel (LCP) to identify the faulty bridge. - Check pulses issued by TTM211 module and pulses at the TPT214 input. CAUTION: On another pulse fault, the unit will trip. Note: Pulse fault may be indicated on cards LED before excitation starting (no pulse due to phase reference missing on TTM211 module)
106: Excitation cubicle door opd	One of excitation door is opened. Detection: Excitation cubicle door contact. Consequence: Alarm. Blocking of field breaker closing command.	Close the excitation cubicle doors before field breaker closing. Check pulses issued by TTM211 module and pulses at the TPT214 input.
107: MX3 protection watchdog	MX3 protection watchdog is lost Detection: MX3 protection contact. Consequence: Trip request. Inverter forcing and field breaker opening after the unit breaker is opened.	The excitation MX3 protection is lost. Check operation, power supply of the protection.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
108: Distributed IO fault	Loss of distributed IO control. Detection: Regulation software detection. Consequence: Alarm. Regulator changeover.	• Communication towards the distributed IO has been lost. – Check distributed IO modules LEDs. – Check connection between IO blocs. Note: it is necessary to power down and up the IO blocks after repair. Note: Closing the field breaker will not be possible unless the default is repaired.
109: Homolog Distributed IO fault	Loss of distributed IO control on homolog regulator. Detection: Regulation software detection. Consequence: Alarm. Homolog regulator not available.	• Communication towards the other regulator for distributed IO has been lost. – Check connection between distributed IO and homolog regulator.
110: Regulator battery low	Battery of regulator is low. Detection: Regulation software detection. Consequence: Possible lost of programs on power supply lost.	• Battery of regulator is low. Exchange the battery while the regulator has power supply.
111: Unit fault	The excitation cubicle has been tripped by a unit fault. Detection: Regulation software detection. Consequence: Tripping of the excitation field breaker.	• Check the reason of the unit fault (external to the excitation system).

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
112: Pulse fault detection inhibition	The pulse fault function is inhibited. Detection: Regulation software detection. Consequence: Pulse amplifiers modules are invalidated and no pulse is generated.	• This operation is usually dedicated for testing purposes. • Valid the pulse fault detection on the LCP, in AVR parameters setting menu, TEST menu, NEXT page.
113: FIP medium fault	One FIP disconnection is detected on regulators. Detection: Regulation software detection. Consequence: Loss on F8000 redundancy.	• Check connection of FIP medium on both regulators.
114: FIP fault	FIP disconnection is detected on regulators. Detection: Regulation software detection. Consequence: No possibility to perform a regulator changeover command.	• Check connection of FIP medium on both regulators.
115 :IO forced	IO has been forced. Detection: Regulation software detection. Consequence: Possible consequences on the process depending on the forced values.	• Unforce the forced variables with controcad or controcad loader.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
116 : Distributed IO fuse fault	Fuse is blown on Distributed IO. Detection: Regulation software detection. Consequence: IO are not operative any more. Trip request.	• Check the inputs/outputs on the distributed IO modules. • Check the distributed IO modules. • Change the fuse.
117 :Excit transf temperature sensor flt	Data coming from one or several sensor of the excitation transformer is incorrect. Detection: Regulation software detection. Consequence: Alarm. Impossibility to close the field breaker.	• Refer to the faulty transducer (alarms 119, 120, 121).
118 :Excit transf temperature sensor 1 flt	Data coming from sensor 1 of the excitation transformer is incorrect. Detection: Regulation software detection. Consequence: Alarm. Impossibility to close the field breaker.	• Check the RTD 1 coming from the excitation transformer. Exchange it if necessary. • Check the distributed IO sensor module.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
119 :Excit transf temperature sensor 2 flt	Data coming from sensor 2 of the excitation transformer is incorrect. Detection: Regulation software detection. Consequence: Alarm. Impossibility to close the field breaker.	• Check the RTD 2 coming from the excitation transformer. Exchange it if necessary. • Check the distributed IO sensor module.
120 :Excit transf temperature sensor 3 flt	Data coming from sensor 3 of the excitation transformer is incorrect. Detection: Regulation software detection. Consequence: Alarm. Impossibility to close the field breaker.	• Check the RTD 3 coming from the excitation transformer. Exchange it if necessary. • Check the distributed IO sensor module.
121: AC Feeder Fault	Faulty AC power supply. Detection: Fuses on AC power supply. Consequence: Alarm.	• Power supply fault detected by fuses. – Check the AC power supply circuit.
122: Trip on TPC fault	Trip on TPC fault. Detection: Refer to TPC215 user manual Consequence:	• Refer to TPC215 user manual.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
123: Change over on TPC demand	Change over on TPC demand. Detection: Refer to TPC215 user manual Consequence:	Refer to TPC215 user manual.
124: Regulator changeover on pulse fault	Regulator changeover on pulse fault. Detection: Refer to TPC215 user manual Consequence:	Refer to TPC215 user manual.
125: TPC1 inhibited	TPC1 inhibited. Detection: Refer to TPC215 user manual Consequence:	Refer to TPC215 user manual.
126: TPC2 inhibited	Same as alarm 125 on TPC 2.	Same as alarm 125 on TPC 2
127: TPC3 inhibited	Same as alarm 125 on TPC 3.	Same as alarm 125 on TPC 3.
128: TPC4 inhibited	Same as alarm 125 on TPC 4.	Same as alarm 125 on TPC 4.
129: TPC5 inhibited	Same as alarm 125 on TPC 5.	Same as alarm 125 on TPC 5.
130: TPC6 inhibited	Same as alarm 125 on TPC 6.	Same as alarm 125 on TPC 6.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
131: Rotating diode fault	Rotating diode fault. Detection: MDEX1 (IEX164 card) output. Consequence: Trip	• A exciter rotating diode have been detected – Check the brushless exciter rotating diodes
132: TPC non blocking alarm	TPC non blocking alarm. Detection: Refer to TPC215 user manual Consequence:	• Refer to TPC215 user manual.
133: Manu underexcitation limitation active Manu underexcitation limitation active	The underexcitation limit value has been reached in manual mode. Detection: Regulation software detection. Consequence: Reactive power or excitation current minimum limitation	- Increase the excitation current setpoint.
134: Rotor voltage measure fault	Rotor voltage measurement fault. Detection: Rotor voltage measurement. Consequence: Invalid rotor voltage measurement. Regulator change over (if possible). Rotor temperature calculation unavailable.	• Rotor voltage measurement fault – Check the rotor voltage transducer. – Check the rotor voltage measurement circuit. – Check the rotor current measurement module.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
144: Braking Sequence Faulty	Braking Sequence Fault. Detection: Braking Sequence activation too long Braking sequence time exceeded Service mode condition not ready Consequence: Alarm Tripping of the excitation field breaker.	• Disconnecter Switch not closing after braking sequence initialized. – Check disconnecter switch functionality. • Field Breaker open demand by speed (<1%) not received after the preset braking time frame (10 mins) exceeded. – Check digital input of Field breaker open demand by speed. • Contactor changeover from Q06 to Q05 does not take place for normal mode operation preparation after field breaker open demand by speed signal is received. – Check contactor control circuit. – Check field breaker open demand by speed external signal
145: Braking Sequence Activation Fault	Braking Sequence Activation Fault. Detection: Braking Sequence activation too long Consequence: Alarm Tripping of the excitation field breaker.	• Disconnecter Switch not closing after braking sequence initialized. – Check disconnecter switch functionality.
146: Braking Time Exceeded Fault	Braking Time Exceeded Fault. Detection: Braking sequence time exceeded Consequence: Alarm Tripping of the excitation field breaker.	• Field Breaker open demand by speed (<1%) not received after the preset braking time frame (10 mins) exceeded. – Check digital input of Field breaker open demand by speed.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
147: Normal Mode Not Ready	Service mode condition not ready Detection: Service mode condition not ready Consequence: Not able to start excitation system	• Contactor changeover from Q06 to Q05 does not take place for normal mode operation preparation after field breaker open demand by speed signal is received. – Check contactor control circuit. – Check field breaker open demand by speed external signal. * Make sure the BRAKING MODE SELECTION (external digital input signal) is reset to normal mode after braking sequence to reset the system to normal mode and also to reset all faulty braking sequence alarm before restart the generator.

5.5.2 Bridge's alarms

The following section details the bridge's fault.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
400: Fan 1 fuse fault - Bridge 1	A fault has been detected on the cooling fan 1 of thyristor bridge 1 fuse. Detection: Fuse contact. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	• Fault on a fuse of the fan 1 of bridge 1. – Check the relevant cooling fan motor and the cooling command sequence in the excitation cubicle. – Check the wiring and the transformer for a short circuit. – Change the faulty fuses. – Check and reset the overload relay. CAUTION: A fault in another bridge will cause the unit to trip.
401: Fan 1 thermal relay flt - Bridge 1	A fault has been detected on the cooling fan 1 of a thyristor bridge 1 thermal relays. Thermal protection threshold: refer to the Preventive maintenance section. Detection: Thermal relay. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	• Fault on thermal relay of the fan 1 of bridge 1. – Check the relevant cooling fan motor and the cooling command sequence in the excitation cubicle. – Check the wiring and the transformer for a short circuit. – Check and reset the overload relay. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
402: Fan 2 fuse fault - Bridge 1	A fault has been detected on the cooling fan 2 of thyristor bridge 1 fuse. Detection: Fuse contact. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	• Fault on a fuse of the fan 2 of bridge 1. – Check the relevant cooling fan motor and the cooling command sequence in the excitation cubicle. – Check the wiring and the transformer for a short circuit. – Change the faulty fuses. – Check and reset the overload relay. CAUTION: A fault in another bridge will cause the unit to trip.
403: Fan 2 thermal relay flt - Bridge 1	A fault has been detected on the cooling fan 2 of a thyristor bridge 1 thermal relays. Thermal protection threshold: refer to the Preventive maintenance section. Detection: Thermal relay. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	• Fault on thermal relay of the fan 2 of bridge 1. – Check the relevant cooling fan motor and the cooling command sequence in the excitation cubicle. – Check the wiring and the transformer for a short circuit. – Check and reset the overload relay. CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
404: Thyristor Bridge 1 Fuses	A fuse is blown in thyristor bridge 1. Detection: Fuse blowing detector in bridge 1. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	- Faulty fuse in thyristor bridge 1 of the excitation cubicle - Check the thyristors of the bridge. Change the faulty thyristors and fuses. - Faulty circuit downstream from the thyristor. - Check the downstream circuit. CAUTION: A fault in another bridge may cause the unit to trip.
405: Thyristor Bridge 1 Over-temp.	A temperature fault has been detected on thyristor bridge 1. Upper threshold: refer to the Preventive maintenance section. Detection: Thermostat on top of the bridge, or temperature switch on the heat sink. Consequence: Inhibition of firing pulses to the thyristor bridge 1: operation with the other bridge(s).	- Temperature fault in a thyristor bridge 1 of the excitation cubicle. - Check the cooling air inlet filters and the air outlet. - Temperature fault due to overloading of one phase. - Check the current sharing between the bridges. - Check all connections of the bridge (internal and external busbars). CAUTION: A fault in another bridge will cause the unit to trip.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
406: Pulse amplifier bridge 1 Fault	Fault on TPA213 module of bridge 1. Detection: MOK signal from the TPA213 module. Consequence: Alarm. Bridge 1 blocking. Bridge changeover (TWIN or redundant TWIN configuration).	- A TPA213 module has been detected faulty on bridge 1. - Check the TPA213 power supplies of bridge 1. CAUTION: On fault of another TPA213 module fault, the unit will trip.
407: Ventilation flt bridge 1	Ventilation fault on bridge 1. Detection: Pressure detector on bridge 1. Consequence: Alarm. Bridge 1 blocking. Bridge changeover (TWIN or redundant TWIN configuration).	- A ventilation fault has been detected on bridge 1. - Check operation of the fan and rotating direction. - Check the corresponding pressure detector. CAUTION: If we have this alarm The unit will trip. In the twin or redundant configuration we have this alarm when we have lost the two fans .
408: Pulse flt bridge 1	Pulses are missing on bridge 1. Detection: TPA213 pulse fault output of bridge 1. Consequence: Alarm. Bridge 1 blocking. Bridge changeover (TWIN or redundant TWIN configuration).	- A pulse fault has been detected by the TPA213 module on bridge 1. - Check pulses issued by TTM211 module and pulses at the TPT214 input of bridge 1. CAUTION: On another pulse fault, the unit will trip. Note: Pulse fault may be indicated on cards LED before excitation starting (no pulse due to reference phase missing on TTM211 module)

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
409: Clipper Fuse Fault Bridge 1	A fuse has been detected as faulty in the overvoltage protection device of bridge 1. Detection: Fuse. Consequence: Alarm.	- Short-circuited component in the overvoltage protection device of bridge 1. - Check the diode bridge and the RC circuit when the system is stopped. Note: Closing the field breaker will not be possible unless the default is repaired.
410: Thyristor 1 Bridge 1 conduction flt	Non conduction of thyristor No. 1 in rectifier bridge 1. Detection: Hall-effect sensor in the AC power supply to the thyristor bridge 1. Consequence: Alarm.	- Thyristor No. 1 of a thyristor bridge 1 in the excitation cubicle is faulty. - Check wiring from the pulse generator circuit. - Check the gate circuit components. - Check the busbar connections of the bridge. - Check the thyristors of the bridge. CAUTION: A fault in another bridge will cause the unit to trip.
411: Thyristor 2 Bridge 1 conduction flt	Same as alarm 410 on thyristor 2 bridge 1.	Same as alarm 410 on thyristor 2 bridge 1.
412: Thyristor 3 Bridge 1 conduction flt	Same as alarm 410 on thyristor 3 bridge 1.	Same as alarm 410 on thyristor 3 bridge 1.
413: Thyristor 4 Bridge 1 conduction flt	Same as alarm 410 on thyristor 4 bridge 1.	Same as alarm 410 on thyristor 4 bridge 1.
414: Thyristor 5 Bridge 1 conduction flt	Same as alarm 410 on thyristor 5 bridge 1.	Same as alarm 410 on thyristor 5 bridge 1.
415: Thyristor 6 Bridge 1 conduction flt	Same as alarm 410 on thyristor 6 bridge 1.	Same as alarm 410 on thyristor 6 bridge 1.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
416: Fan 1 fuse fault - Bridge 2	Same as alarm 400 on bridge 2.	Same as alarm 400 on bridge 2
417: Fan 1 thermal relay flt - Bridge 2	Same as alarm 401 on bridge 2.	Same as alarm 401 on bridge 2
418: Fan 2 fuse fault - Bridge 2	Same as alarm 402 on bridge 2.	Same as alarm 402 on bridge 2
419: Fan 2 thermal relay flt - Bridge 2	Same as alarm 403 on bridge 2.	Same as alarm 403 on bridge 2
420: Thyristor Bridge 2 Fuses	Same as alarm 404 on bridge 2.	Same as alarm 404 on bridge 2
421: Thyristor Bridge 2 Over-temp.	Same as alarm 405 on bridge 2.	Same as alarm 405 on bridge 2
422: Pulse amplifier bridge 2 Fault	Same as alarm 406 on bridge 2.	Same as alarm 406 on bridge 2
423: Ventilation flt bridge 2	Same as alarm 407 on bridge 2.	Same as alarm 407 on bridge 2
424: Pulse flt bridge 2	Same as alarm 408 on bridge 2.	Same as alarm 408 on bridge 2
425: Clipper Fuse Fault Bridge 2	Same as alarm 409 on bridge 2.	Same as alarm 409 on bridge 2
426: Thyristor 1 Bridge 2 conduction flt	Same as alarm 410 on bridge 2.	Same as alarm 410 on bridge 2
427: Thyristor 2 Bridge 2 conduction flt	Same as alarm 411 on bridge 2.	Same as alarm 411 on bridge 2
428: Thyristor 3 Bridge 2 conduction flt	Same as alarm 412 on bridge 2.	Same as alarm 412 on bridge 2
429: Thyristor 4 Bridge 2 conduction flt	Same as alarm 413 on bridge 2.	Same as alarm 413 on bridge 2

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
430: Thyristor 5 Bridge 2 conduction flt	Same as alarm 414 on bridge 2.	Same as alarm 414 on bridge 2
431: Thyristor 6 Bridge 2 conduction flt	Same as alarm 415 on bridge 2.	Same as alarm 415 on bridge 2
432: Fan 1 fuse fault - Bridge 3	Same as alarm 400 on bridge 3.	Same as alarm 400 on bridge 3
433: Fan 1 thermal relay flt - Bridge 3	Same as alarm 401 on bridge 3.	Same as alarm 401 on bridge 3
434: Fan 2 fuse fault - Bridge 3	Same as alarm 402 on bridge 3.	Same as alarm 402 on bridge 3
435: Fan 2 thermal relay flt - Bridge 3	Same as alarm 403 on bridge 3.	Same as alarm 403 on bridge 3
436: Thyristor Bridge 3 Fuses	Same as alarm 404 on bridge 3.	Same as alarm 404 on bridge 3
437: Thyristor Bridge 3 Over-temp.	Same as alarm 405 on bridge 3.	Same as alarm 405 on bridge 3
438: Pulse amplifier bridge 3 Fault	Same as alarm 406 on bridge 3.	Same as alarm 406 on bridge 3
439: Ventilation flt bridge 3	Same as alarm 407 on bridge 3.	Same as alarm 407 on bridge 3
440: Pulse flt bridge 3	Same as alarm 408 on bridge 3.	Same as alarm 408 on bridge 3
441: Clipper Fuse Fault Bridge 3	Same as alarm 409 on bridge 3.	Same as alarm 409 on bridge 3
442: Thyristor 1 Bridge 3 conduction flt	Same as alarm 410 on bridge 3.	Same as alarm 410 on bridge 3
443: Thyristor 2 Bridge 3 conduction flt	Same as alarm 411 on bridge 3.	Same as alarm 411 on bridge 3

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
444: Thyristor 3 Bridge 3 conduction flt	Same as alarm 412 on bridge 3.	Same as alarm 412 on bridge 3
445: Thyristor 4 Bridge 3 conduction flt	Same as alarm 413 on bridge 3.	Same as alarm 413 on bridge 3
446: Thyristor 5 Bridge 3 conduction flt	Same as alarm 414 on bridge 3.	Same as alarm 414 on bridge 3
447: Thyristor 6 Bridge 3 conduction flt	Same as alarm 415 on bridge 3.	Same as alarm 415 on bridge 3
448: Fan 1 fuse fault - Bridge 4	Same as alarm 400 on bridge 4.	Same as alarm 400 on bridge 4
449: Fan 1 thermal relay flt - Bridge 4	Same as alarm 401 on bridge 4.	Same as alarm 401 on bridge 4
450: Fan 2 fuse fault - Bridge 4	Same as alarm 402 on bridge 4.	Same as alarm 402 on bridge 4
451: Fan 2 thermal relay flt - Bridge 4	Same as alarm 403 on bridge 4.	Same as alarm 403 on bridge 4
452: Thyristor Bridge 4 Fuses	Same as alarm 404 on bridge 4.	Same as alarm 404 on bridge 4
453: Thyristor Bridge 4 Over-temp.	Same as alarm 405 on bridge 4.	Same as alarm 405 on bridge 4
454: Pulse amplifier bridge 4 Fault	Same as alarm 406 on bridge 4.	Same as alarm 406 on bridge 4
455: Ventilation flt bridge 4	Same as alarm 407 on bridge 4.	Same as alarm 407 on bridge 4
456: Pulse flt bridge 4	Same as alarm 408 on bridge 4.	Same as alarm 408 on bridge 4
457: Clipper Fuse Fault Bridge 4	Same as alarm 409 on bridge 4.	Same as alarm 409 on bridge 4

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
458: Thyristor 1 Bridge 4 conduction flt	Same as alarm 410 on bridge 4.	Same as alarm 410 on bridge 4
459: Thyristor 2 Bridge 4 conduction flt	Same as alarm 411 on bridge 4.	Same as alarm 411 on bridge 4
460: Thyristor 3 Bridge 4 conduction flt	Same as alarm 412 on bridge 4.	Same as alarm 412 on bridge 4
461: Thyristor 4 Bridge 4 conduction flt	Same as alarm 413 on bridge 4.	Same as alarm 413 on bridge 4
462: Thyristor 5 Bridge 4 conduction flt	Same as alarm 414 on bridge 4.	Same as alarm 414 on bridge 4
463: Thyristor 6 Bridge 4 conduction flt	Same as alarm 415 on bridge 4.	Same as alarm 415 on bridge 4
464: Fan 1 fuse fault - Bridge 5	Same as alarm 400 on bridge 5.	Same as alarm 400 on bridge 5
465: Fan 1 thermal relay flt - Bridge 5	Same as alarm 401 on bridge 5.	Same as alarm 401 on bridge 5
466: Fan 2 fuse fault - Bridge 5	Same as alarm 402 on bridge 5.	Same as alarm 402 on bridge 5
467: Fan 2 thermal relay flt - Bridge 5	Same as alarm 403 on bridge 5.	Same as alarm 403 on bridge 5
468: Thyristor Bridge 5 Fuses	Same as alarm 404 on bridge 5.	Same as alarm 404 on bridge 5
469: Thyristor Bridge 5 Over-temp.	Same as alarm 405 on bridge 5.	Same as alarm 405 on bridge 5
470: Pulse amplifier bridge 5 Fault	Same as alarm 406 on bridge 5.	Same as alarm 406 on bridge 5

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
471: Ventilation flt bridge 5	Same as alarm 407 on bridge 5.	Same as alarm 407 on bridge 5
472: Pulse flt bridge 5	Same as alarm 408 on bridge 5.	Same as alarm 408 on bridge 5
473: Clipper Fuse Fault Bridge 5	Same as alarm 409 on bridge 5.	Same as alarm 409 on bridge 5
474: Thyristor 1 Bridge 5 conduction flt	Same as alarm 410 on bridge 5.	Same as alarm 410 on bridge 5
475: Thyristor 2 Bridge 5 conduction flt	Same as alarm 411 on bridge 5.	Same as alarm 411 on bridge 5
476: Thyristor 3 Bridge 5 conduction flt	Same as alarm 412 on bridge 5.	Same as alarm 412 on bridge 5
477: Thyristor 4 Bridge 5 conduction flt	Same as alarm 413 on bridge 5.	Same as alarm 413 on bridge 5
478: Thyristor 5 Bridge 5 conduction flt	Same as alarm 414 on bridge 5.	Same as alarm 414 on bridge 5
479: Thyristor 6 Bridge 5 conduction flt	Same as alarm 415 on bridge 5.	Same as alarm 415 on bridge 5
480: Fan 1 fuse fault - Bridge 6	Same as alarm 400 on bridge 6.	Same as alarm 400 on bridge 6
481: Fan 1 thermal relay flt - Bridge 6	Same as alarm 401 on bridge 6.	Same as alarm 401 on bridge 6
482: Fan 2 fuse fault - Bridge 6	Same as alarm 402 on bridge 6.	Same as alarm 402 on bridge 6
483: Fan 2 thermal relay flt - Bridge 6	Same as alarm 403 on bridge 6.	Same as alarm 403 on bridge 6
484: Thyristor Bridge 6 Fuses	Same as alarm 404 on bridge 6.	Same as alarm 404 on bridge 6

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
485: Thyristor Bridge 6 Over-temp.	Same as alarm 405 on bridge 6.	Same as alarm 405 on bridge 6
486: Pulse amplifier bridge 6 Fault	Same as alarm 406 on bridge 6.	Same as alarm 406 on bridge 6
487: Ventilation flt bridge 6	Same as alarm 407 on bridge 6.	Same as alarm 407 on bridge 6
488: Pulse flt bridge 6	Same as alarm 408 on bridge 6.	Same as alarm 408 on bridge 6
489: Clipper Fuse Fault Bridge 6	Same as alarm 409 on bridge 6.	Same as alarm 409 on bridge 6
490: Thyristor 1 Bridge 6 conduction flt	Same as alarm 410 on bridge 6.	Same as alarm 410 on bridge 6
491: Thyristor 2 Bridge 6 conduction flt	Same as alarm 411 on bridge 6.	Same as alarm 411 on bridge 6
492: Thyristor 3 Bridge 6 conduction flt	Same as alarm 412 on bridge 6.	Same as alarm 412 on bridge 6
493: Thyristor 4 Bridge 6 conduction flt	Same as alarm 413 on bridge 6.	Same as alarm 413 on bridge 6
494: Thyristor 5 Bridge 6 conduction flt	Same as alarm 414 on bridge 6.	Same as alarm 414 on bridge 6
495: Thyristor 6 Bridge 6 conduction flt	Same as alarm 415 on bridge 6.	Same as alarm 415 on bridge 6
496: Bridge 1 isolated	Bridge 1 has been isolated Detection: Isolator position. Consequence: Bridge fault.	Close the isolator of the bridge 1 after having cleared the faults on the bridge.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
497: Bridge 2 isolated	Same as alarm 496 on bridge 2.	• Same as alarm 496 on bridge 2
498: Bridge 3 isolated	Same as alarm 496 on bridge 3.	• Same as alarm 496 on bridge 3
499: Bridge 4 isolated	Same as alarm 496 on bridge 4.	• Same as alarm 496 on bridge 4
500: Bridge 5 isolated	Same as alarm 496 on bridge 5.	• Same as alarm 496 on bridge 5
501: Bridge 6 isolated	Same as alarm 496 on bridge 6.	• Same as alarm 496 on bridge 6
504: Pressure fault Bridge 1	Pressure fault Bridge 1. Detection: Pressure detector on bridge 1. Consequence: Alarm. Bridge 1 blocking. Bridge changeover (TWIN or redundant TWIN configuration).	• A ventilation fault has been detected on bridge 1. – Check operation of the fan and rotating direction. – Check the corresponding pressure detector. • CAUTION: On another ventilation fault (alarm 407), the unit will trip.
505: Pressure fault Bridge 2	Same as alarm 504 on bridge 2.	• Same as alarm 504 on bridge 2.
506: Pressure fault Bridge 3	Same as alarm 504 on bridge 3.	• Same as alarm 504 on bridge 3.
507: Pressure fault Bridge 4	Same as alarm 504 on bridge 4.	• Same as alarm 504 on bridge 4.
508: Pressure fault Bridge 5	Same as alarm 504 on bridge 5.	• Same as alarm 504 on bridge 5.
509: Pressure fault Bridge 6	Same as alarm 504 on bridge 6.	• Same as alarm 504 on bridge 6.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
512: TPC 1 non blocking alarm	TPC 1 non blocking alarm. Detection: Refer to TPC215 user manual Consequence:	Refer to TPC215 user manual.
513: TPC 2 non blocking alarm	Same as alarm 512 on TPC 2	Same as alarm 512 on TPC 2
514: TPC 3 non blocking alarm	Same as alarm 512 on TPC 3	Same as alarm 512 on TPC 3
515: TPC 4 non blocking alarm	Same as alarm 512 on TPC 4	Same as alarm 512 on TPC 4
516: TPC 5 non blocking alarm	Same as alarm 512 on TPC 5	Same as alarm 512 on TPC 5
517: TPC 6 non blocking alarm	Same as alarm 512 on TPC 6	Same as alarm 512 on TPC 6

5.5.3 TTM211 alarms

The following section tails the TTM211 card fault.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
600: TTM blocking alarm	A blocking alarm has been detected in the TTM211 module Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- A blocking alarm has been detected by TTM211 module. Refer to the specific fault type indicated on the Local Control Panel (LCP) or to the maintenance computer indication to identify the fault type: 603: TTM setpoint line break 604: TTM setpoint line saturated 606: TTM periods interrupt failure 607: TTM phase reference fault 608: TTM phase order fault CAUTION: A fault in another TTM211 will cause the unit to trip.
601: TTM alarm	A non blocking alarm has been detected in the TTM211 module Detection: TTM211 module. Consequence: None.	- A non blocking alarm has been detected by TTM211 module. Refer to the specific fault type indicated on the Local Control Panel (LCP) or to the maintenance computer to identify the fault type: 605: TTM Measured period < min
603: TTM setpoint line break	The 4-20 mA setpoint is under its minimum value. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- The 4-20 mA setpoint to the TTM211 is detected under its minimum value. This information corresponds to fault CRL described in paragraph 5.4.10. - Check the 4-20 mA input of TTM211. - Check the 4-20 mA output of ALG391. - Exchange the TTM211 (internal converter fault).

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
604: TTM setpoint line saturated	The 4-20 mA setpoint is over its maximum value. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- The 4-20 mA setpoint to the TTM211 is detected over its maximum value. This information corresponds to fault CSAT described in paragraph 5.4.10. - Check the 4-20 mA input of TTM211. - Check the 4-20 mA output of ALG391. - Exchange the TTM211 (internal converter fault).
605: TTM Measured period < min	The period on the synchronising voltage is under its minimum value. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- The period on the synchronising voltage is under its minimum value. This information corresponds to fault PMIN described in paragraph 5.4.10. - Check the synchronising voltage at the TTM211 input. - Check the synchronising voltage transformer. - Check the synchronising voltage fuses.
606: TTM periods interrupt failure	The period on the synchronising voltage is abnormal. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- The period on the synchronising voltage is abnormal. This information corresponds to fault DPIT described in paragraph 5.4.10. - Check the synchronising voltage at the TTM211 input. - Check the synchronising voltage transformer. - Check the synchronising voltage fuses.
607: TTM phase reference fault	The amplitude on the synchronising voltage is abnormal. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	- The amplitude on the synchronising voltage is abnormal (unbalance detected). This information corresponds to fault DREF described in paragraph 5.4.10. - Check the synchronising voltage at the TTM211 input. - Check the synchronising voltage transformer. - Check the synchronising voltage fuses.

Alarm message	Fault description	(•) POSSIBLE REASONS (-) ACTIONS REQUIRED
608: TTM phase order fault	The phase sequence on the synchronising voltage is abnormal. Detection: TTM211 module. Consequence: - Regulator change over (if possible).	• The phase sequence of the synchronising voltage is abnormal. This information corresponds to fault DOPH described in paragraph 5.4.10. – Check the synchronising voltage at the TTM211 input. – Check the synchronising voltage transformer. – Check the synchronising voltage fuses.
609: TTM parameters invalid	A error has been detected on the TTM211 module when transferring the TTM211 parameters. Detection: TTM211 module. Consequence: - TTM211 erratic parameters.	• Resend and recheck all the parameters of the corresponding TTM211 module.

5.6 Repair procedures

5.6.1 Bus bar mounting torque

Before making a connection, both contact surfaces must be cleaned and degreased with an appropriate solvent.

See the clamping torques of the hexagon head bolts (class 8.8) in the following table:

5.6.2 Thyristor bridge mounting torque

Before making a connection, both contact surfaces must be cleaned and degreased with an appropriate solvent.

See the clamping torques of the hexagon head bolts (class 8.8) in the following board:

TABLE OF THE TIGHTENING TORQUES FOR WIDE THREAD STEEL NUTS AND BOLTS.
USED FOR BOTH ELECTRICAL AND MECHANICAL ASSEMBLY

Diamètre Nominal (mm)	Pas (mm) (Thread)	Section Résist. IIR ² Moyen (mm) (Average resistance section)	Diamètre Perçage Maxi (mm) (Maximum drilling)	Force axiale dans la vis (daN) (Axial force on the bolt)	Couple de serrage à 80% de la limite élastique (mN) Pour vis inox conforme à T.04121-9 graissée ou acier classe 8-8 non graissée <i>Tightening torque to 80% of the elastic limit, m/daN Stainless steel bolts conforming to T.04121-9 (greased) or 8-8 class stelle (ungreased)</i>					
					Ecrou avec rondelle (nut with washer)		Ecrou embase crantée (Notched base plate nut)		Ecrou Nylstop avec rondelle (Nylstop nut with washer)	
					Plate ou grower (flat or grower)	Contact ou Eventail (Contact or external tooth lock)	Acier (steel)	Cuivre ou alu (copper or aluminium)	Plate (flat)	Contact (contact)
3	0.5	5.03	3.4	150	1	1.5	1.5	1.7	1.2	1.7
4	0.7	8.78	4.5	250	2.5	3.5	3.5	4.1	2.8	3.8
5	0.8	14.2	5.5	380	5	7	8	9.3	5.5	7.5
6	1	20.1	6.6	600	8.5	13	13.5	15.5	10	14.5
8	1.25	36.6	9	1000	19	28	25	29	22	31
10	1.5	58	11	1500	37.5	50	50	59	41.5	54
12	1.75	84.3	14	2400	62	75	79	92	70	83
14	2	115	16	3300	98	120	125	146	110	132
16	2	157	18	4500	155	185	180	210	170	200
18	2.5	192	20	5500	210	260	270	310	230	280
20	2.5	245	22	7100	290	370	390	450	330	410

1cm.Kgf≈0.01 m.daN

1m.daN≈1m.kg

5.6.3 Replacing a regulator module

Special tools

No special tools

Safety requirements

The excitation equipment is switched off and earthed.

Preliminary conditions

The power supplies must be switched off before removing or installing a board.

Before replacing an electronic board, it is recommended to check that all boards are properly connected.

Procedure

De-energize the channel to be repaired by opening the micro-circuit-breaker of the supply

Remove the terminal block from the module, if there is one

Remove the faulty module and replace it by another one of the same type

Put the terminal block back

If the regulator module is a CPU, refer to § 5.6.4 "X20 CPU".

If the regulator module is a TTM module, refer to § 5.6.5 "TTM replacement".

If the regulator module is a TPC module, refer to TPC215 user manual PSM82A415022.

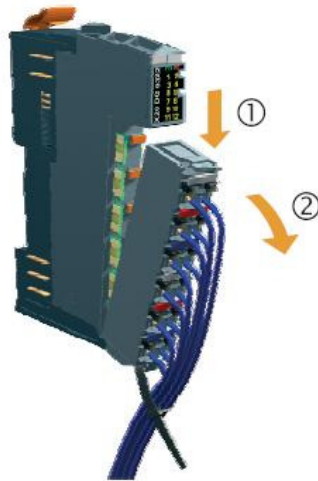
Turn the logic controller back on

5.6.3.1 Removing a terminal block

To remove a terminal block:

Push the locking lever on the terminal block ① to release the terminal block.

Rotate the terminal block down ② until the contacts come free from the module housing.



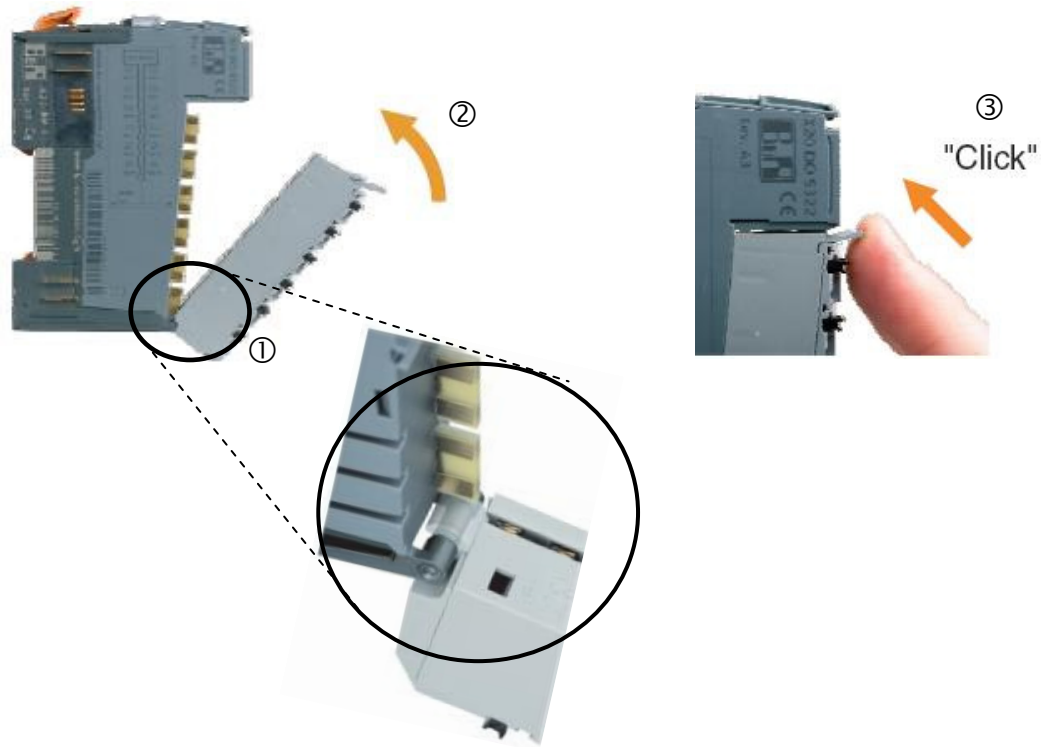
5.6.3.2 Installing a terminal block

To install a terminal block:

Insert the hinge at the bottom of the terminal block into the bus module bottom ①.

Push the terminal block towards the module ② until an audible “click” is heard ③.

This sound indicates that it is in place. If the latch does not catch, the lever must be pushed up.



Caution: Check the labels on the terminal block and the module to make sure that they match. If a terminal block with cables is installed on the wrong module, it may be the subject of certain damage.

5.6.3.3 Removing an electronic module

Use the following procedure to remove a module from its location:

Removing the terminal block (refer to 5.6.3.1- Removing a terminal block).

Locate the locking lever at the top of the module and push it down (towards the module) ①.

While maintaining the pressure, remove the electronic module ②.



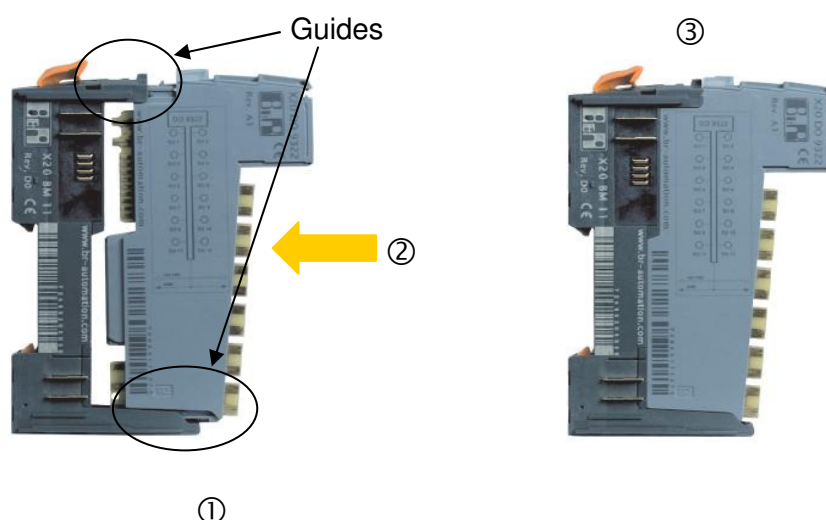
5.6.3.4 Inserting an electronic module

Use the following instructions as a guide for inserting a module in its location on a Bus Module:

Locate the place to install the module. Hold the module firmly towards you (without the terminal block) and the rear towards the bus module.

Insert the electronic module in the guides on the bus module. ①.

Push the electronic module on the bus module ② until the electronic module is locked ③.



5.6.3.5 Removing one or several bus module from the mounting rail

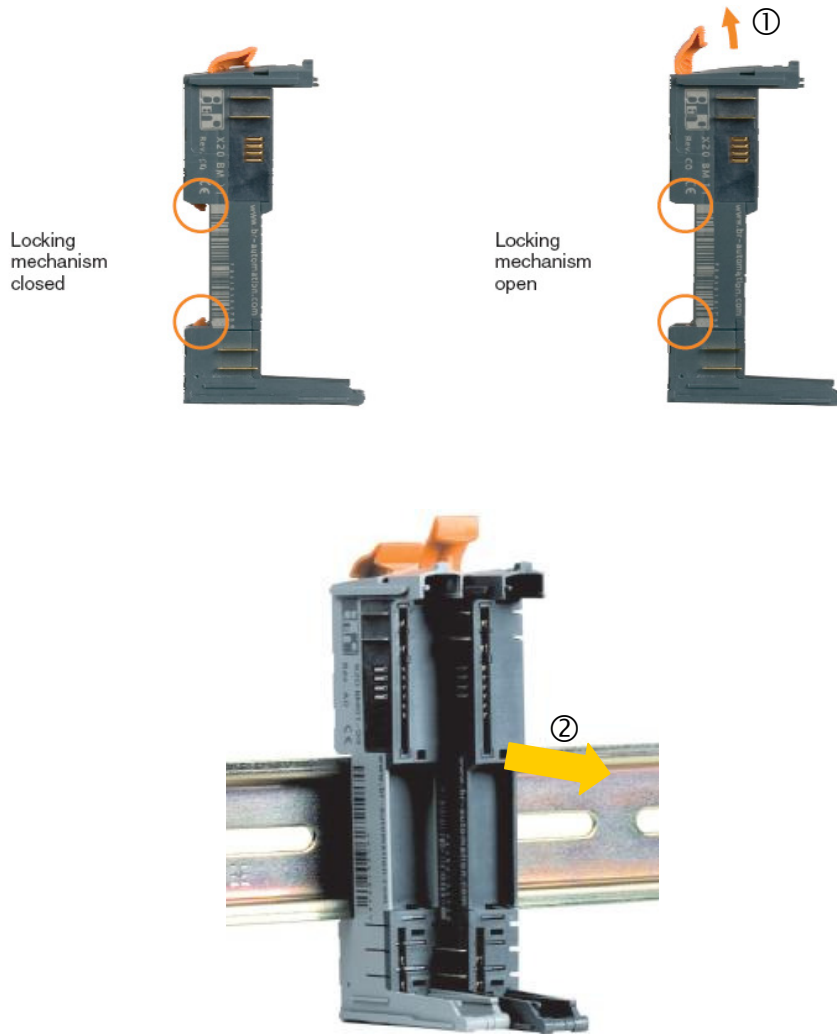
WARNING

Do not insert and do not remove the Bus module when the concerned EPL node is energized. The module could be damaged and persons could be injured.

Use the following procedure to remove one or several Bus modules from their location:

Push the locking lever all the way up on all the modules that you wish to remove from the mounting rail. This opens the locking mechanism for mounting rail installation ①.

Remove the bus module (or bus module block) from the mounting rail. ②



5.6.3.6 Inserting one or several bus modules on the mounting rail

WARNING

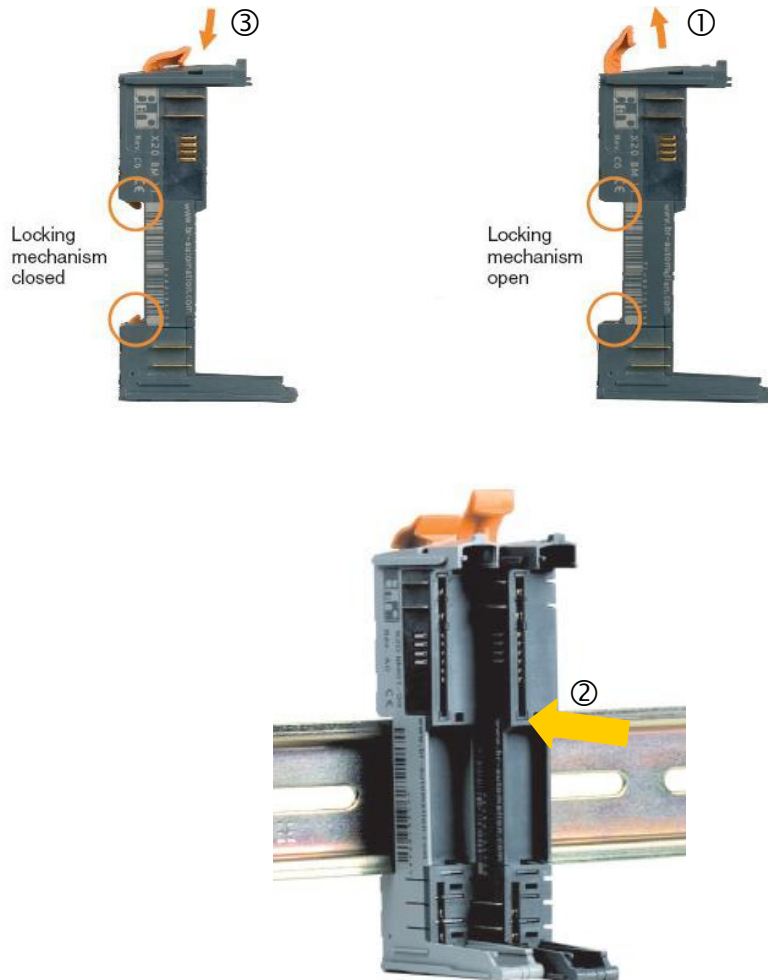
Do not insert and do not remove the Bus module when the concerned EPL node is energized. The module could be damaged and persons could be injured.

Use the following procedure to insert one or several Bus modules to their location:

Push the locking lever all the way up on all the modules that you wish to insert on the mounting rail. This opens the locking mechanism for mounting rail installation①.

Insert the bus module (or bus module block) in the guides of the previously mounted bus module. ②

Slide the bus module in against the mounting rail ② and secure it by pushing down the locking lever ③.

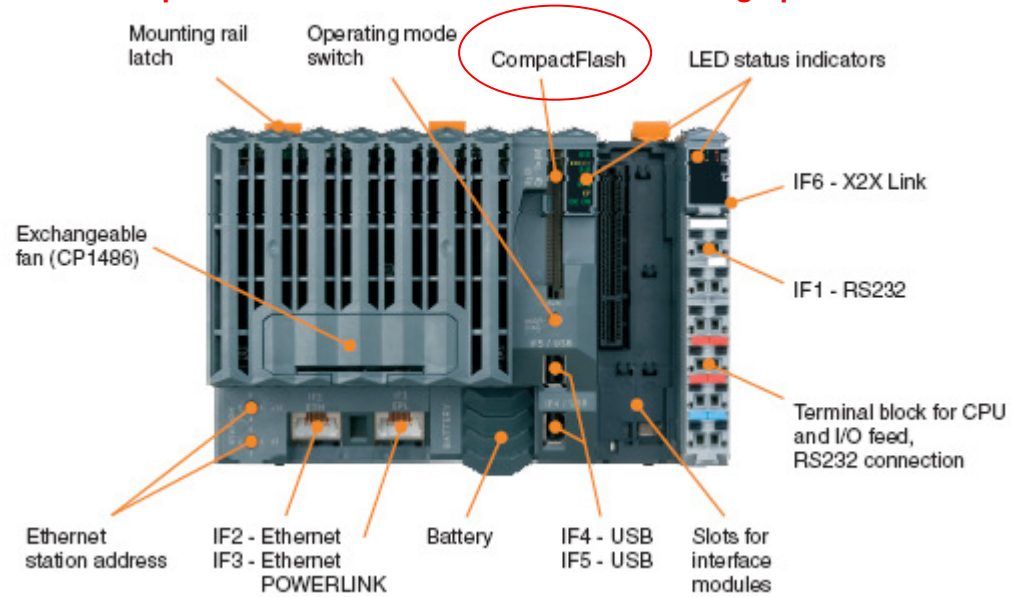


5.6.4 X20 CPU

1. Switch off CPU power supply.

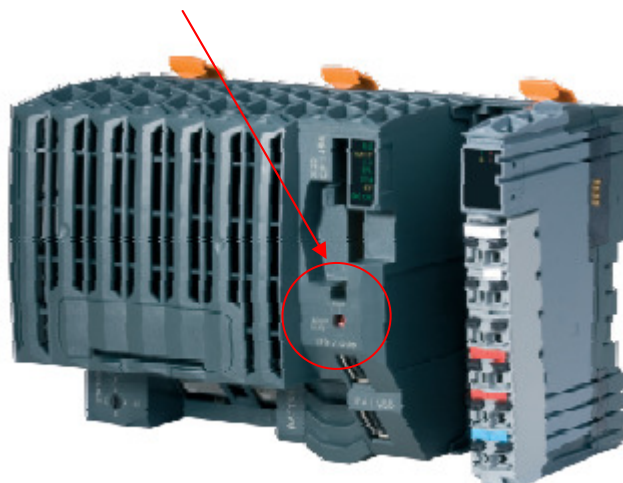
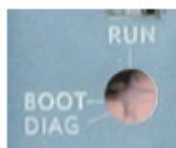
2. Remove compact flash memory. Software is stored in compact flash memory.

Caution: The Compact Flash card cannot be removed during operation.



3. Remove CPU module by pushing the locking lever up.
4. Put the new CPU on the rail and lock it to the rail by pushing the locking lever down. Make sure that connection to the bus is ensured.
5. Put the compact flash memory in the new CPU.
6. Make sure the operating mode switch is in run position.

Operating mode switch



7. Power on the CPU

5.6.5 TTM replacement

Caution: This operation shall not be done while the unit is running.

5.6.5.1 Software used

Controcad , Controcad Loader or Local Control Panel is used to download the parameters of the UMT162 module.

Controcad and Controcad Loader are working with Ethernet communication port of the computer.

Further information is displayed in document "ALSPA AVR commissioning guide" for the TTM parameters downloading.

5.6.5.2 Procedure

Switch off the power supply to the TTM module.

Disconnect cables installed on the TTM module connectors.

Remove the TTM module.

Set up the new TTM module.

Reconnect the cable to the TTM module connectors.

Switch on the power supply to the TTM module.

Check the TTM parameters setting as per the parameters setting specification with Controcad or Controcad loader or the Local Control Panel (Refer to document "ALSPA AVR commissioning guide").

Before restarting the unit, check the correct operation of the TTM module, with the release of the firing pulses on each bridge, and check the right phase and sequence of every thyristor pulses:

Inject voltage at the TTM phase reference voltage transformers. These voltages shall be in the same phase order than those connected to the thyristor bridge.

On Local control Panel force parameter MLOC menu "Local mode control" to 1 and put parameter CLOE to 0% in the TTM211 module . The firing angle on that mode is 90°.

Check that the tested TTM is on the active channel.

Check correct position of the pulse on each thyristor in accordance with injected voltage. Adjust parameter CFI if necessary.

On Local Control Panel force parameter MLOC menu "Local mode control" to 0 in the TTM211 module.

Disconnect the voltage injection on the phase reference transformers.

After restarting, the unit perform a regulator changeover command(for redundant regulators), and check a smooth transfer from one regulator to the other.

5.6.6 TPI163 replacement

Before any replacement of this card, make sure that current measurement circuit is closed.

Check before replacement

Right position of the current rating (1A)

Right position of voltage rating (100V)

5.6.7 PMF175 replacement

To be defined

5.6.8 Replacing a thyristor (TYPE A bridge)

Special tools

Thyristor extractor tool, included with each component

Safety requirements

The thyristor bridge must be earthed before you begin with the maintenance operations.

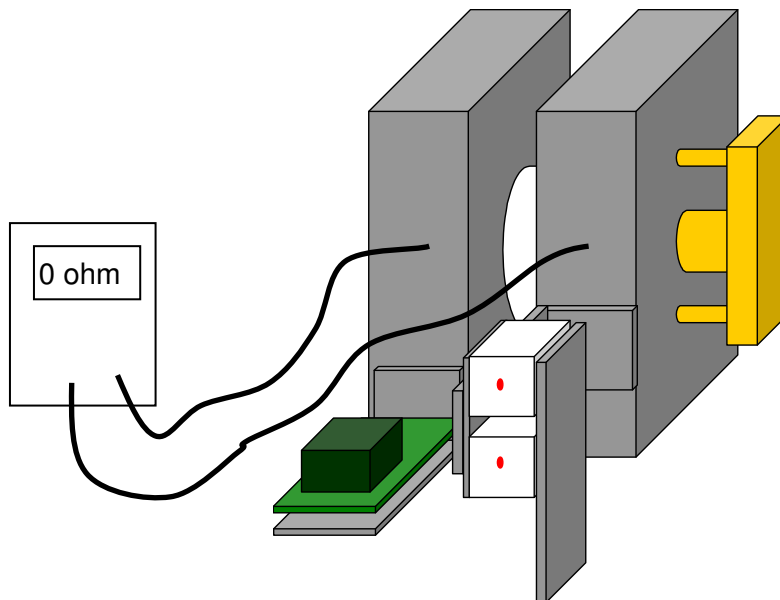
Preliminary conditions

No preliminary works are to be planned.

Procedure

Checking the thyristors

To check a thyristor, it has first to be clamped between two heatsinks .



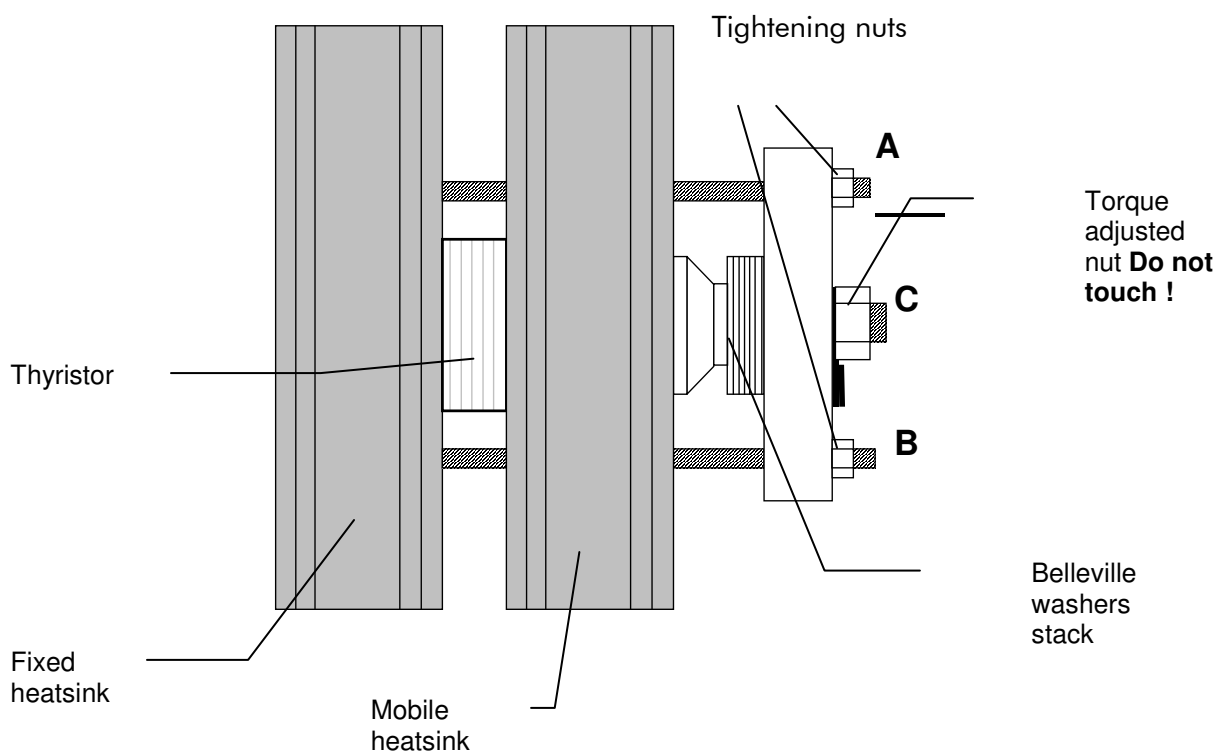
Use an ohmmeter to measure the anode-to-cathode resistance. A thyristor in good condition will have a very high resistance in both directions. The gate-to-cathode resistance, measured in both directions, must be between 10 Ohms and 100 Ohms.

A different resistance indicates a faulty thyristor.

Removing a failed thyristor

It is advised to use components of identical VT range for the legs put in parallel

Respect the tightening torque



Disconnect the gate circuit.

Untighten nuts A & B. You need to untighten these in a equilibrate manner to keep the clamp vertically. For eg, untighten a quarter round on nut A, quarter round on nut B, quarter round on nut A, etc.

Never touch nut C.

Do not remove nuts A & B. just slide them to the end of the screw.

You can now remove the thyristor by pulling it with its holding plastic

If the thyristor doesn't move, you can spread the heaksinks to make it easier. Right heatsink can slide to the right

CAUTION: The central Nilstop nut must not be unscrewed under any circumstances. Doing so would alter the pressure applied by the clamp.

8. Preparation of heatsinks and thyristor

Heatsinks

Clean both surfaces of contact with acetone. This to remove the dried grease that can remain on the heatsinks.

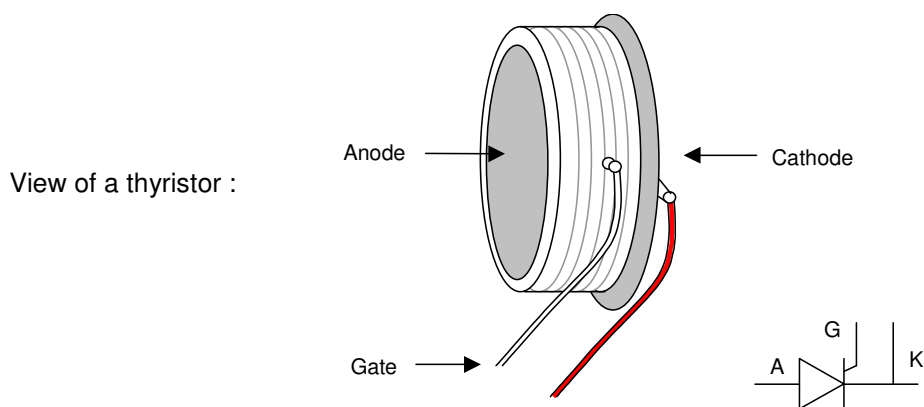
Thyristor

Print on the failed thyristor something like 'defect' with the date of the failure. This to avoid a reuse of faulty thyristors.

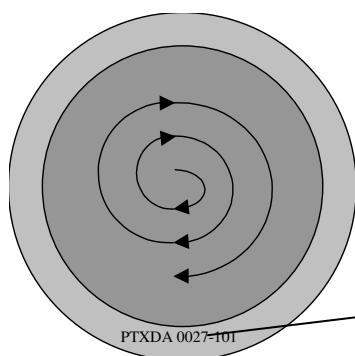
Remove plastic holder, thermal sheath and green gate connector from the old thyristor.

Install on the new thyristor these holder, sheath and connector. Be careful to the right connection of the red and white wire (white = gate, red = cathode). Have a look at others thyristors to be sure.

Warning: When holding the new thyristor with fingers, avoid touching the contact surface. This will create bad oxidation.



Add a slim layer of contact grease on both surfaces of thyristor.



In order to do this, use a lint-free rag and spread the grease in a spiral manner.

The entire surface shall be covered with grease.

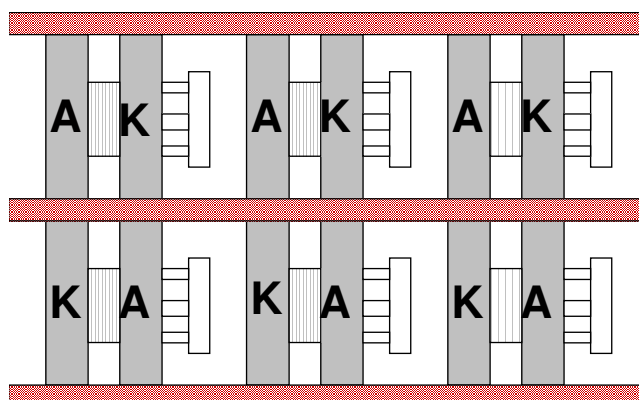
Serial number

Don't forget to note the serial number which is engraved on one face

NB: you can check again the failure of the thyristor with an ohmmeter but you need to put a physical pressure on the thyristor.

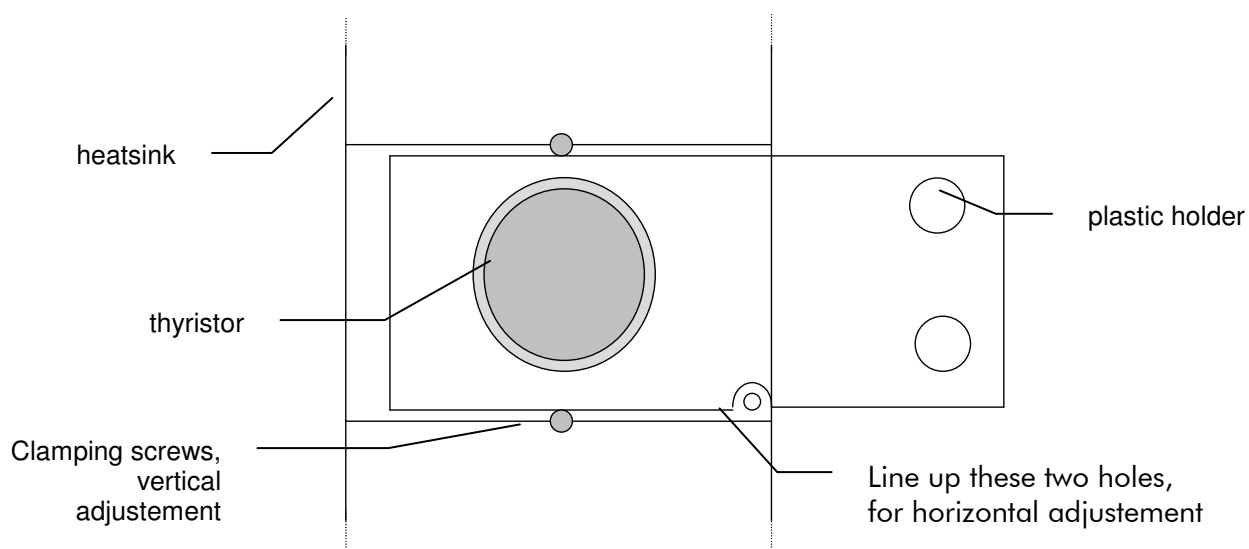
9. Re-installing a thyristor

By holding the thyristor with its plastic holder, insert it between heatsinks. Be careful to the **sense** of the thyristor. Thyristors at the same level shall be in the same sense.



Vertically, the position is automatically adjusted by the two clamp screws.

For horizontal position, you need to line up the hole in the heatsink with the hole in the plastic holder. A white pin fixed in the heatsink hole will help you for this job. If no present, adjust it to have 1mm precision.

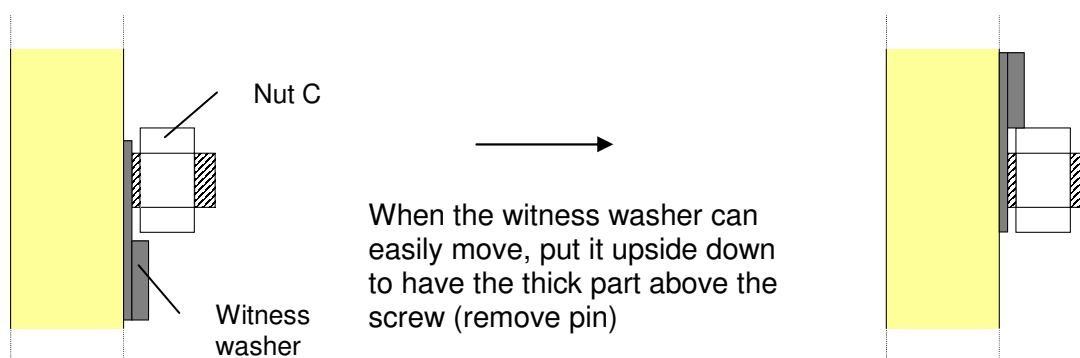


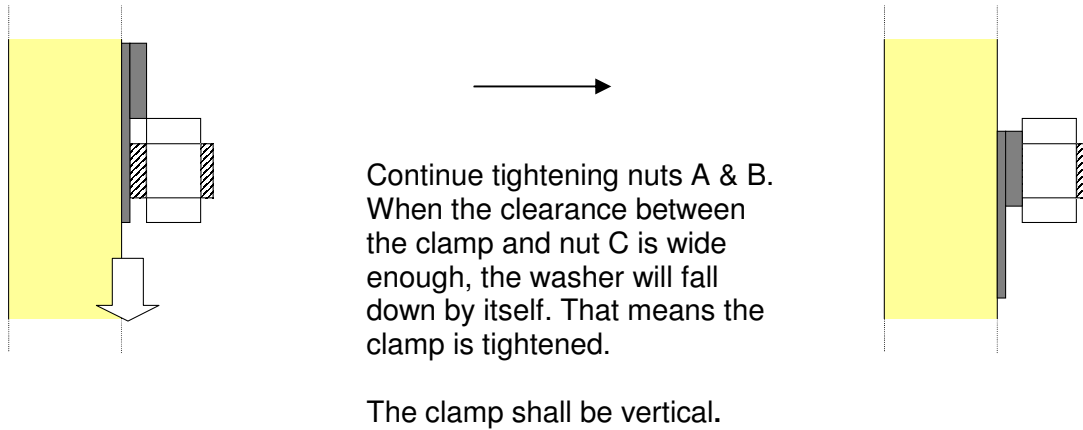
Start to tighten the clamp (nuts A & B) with hand to avoid the thyristor moving.

When the thyristor no more moves, use a 16mm spanner to tighten more the clamp. Make a quarter round on nut A, then on B, go back on A, etc.

At any time you shall check the vertical position of the clamp. If the vertical position is not correct, adjust it by tightening more or less nut A & B.

When witness washer between nut C and clamp is no more squeezed and can be moved follow the procedure below :





Then put back the witness washer in its original position (thick part below)

Reconnect now the thyristor on the gate circuit card.

CAUTION: Insufficient or excessive clamping pressure may lead to destruction of the component.

5.6.9 Replacing a thyristor fuse (TYPE A bridge)

Special tools

Screwdriver

Safety requirements

The excitation equipment is switched off and earthed.

Preliminary conditions

No preliminary works are necessary.

Procedure

It is advised to use components of identical VT range for the legs put in parallel
Respect the tightening torque

Remove the micro-switch from the fuse using the cross recess screw (item 1).

Keep the spindle in position using a male wrench and unscrew the nuts on either side of the fuse (item 2).

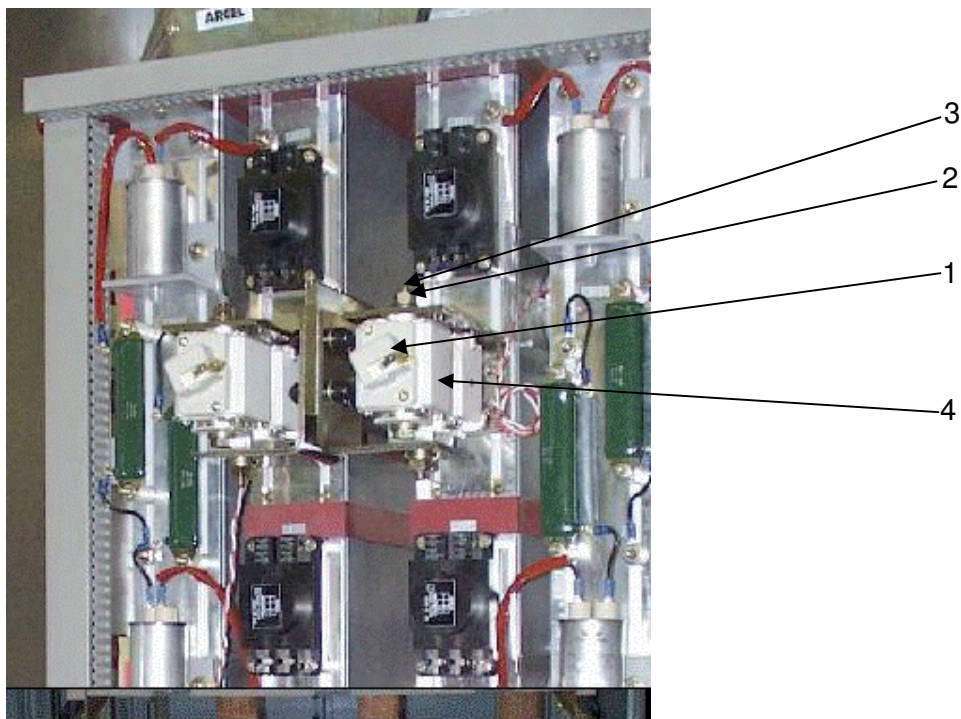
Remove the 2 spindles from the fuse with the male wrench (item 3).

Replace the faulty fuse (item 4).

Tighten the spindles and the nuts equally. Recommended clamping torques are:
13.5 Nm for fuses sized 70-71
15 Nm for fuses sized 72-73

Mount the micro-switch back on the fuse.

CAUTION: Check that no metallic parts are left in the thyristor bridge, since this may cause heavy damage.



5.6.10 Replacing the fan motor (TYPE A bridge)

Special tools

Wrench

Safety requirements

No preliminary works are necessary.

Preliminary conditions

The excitation equipment is switched off and earthed.

Procedure

Remove all fixing bolts from the fan assembly (item 1).

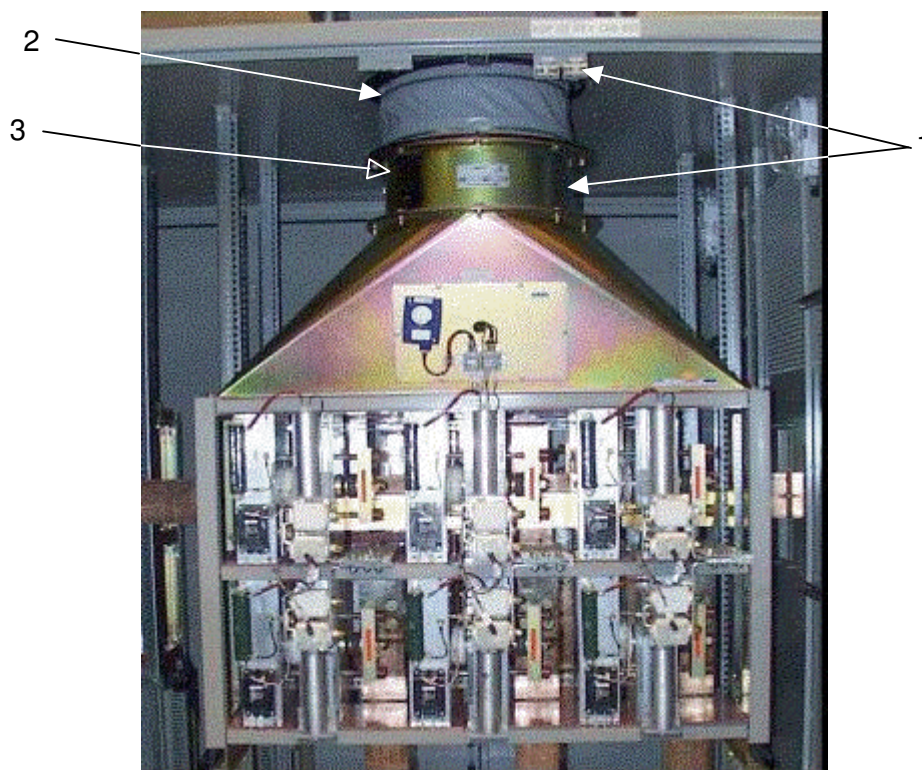
Remove the flexible duct (item 2).

Remove the motor from the flange (item 3).

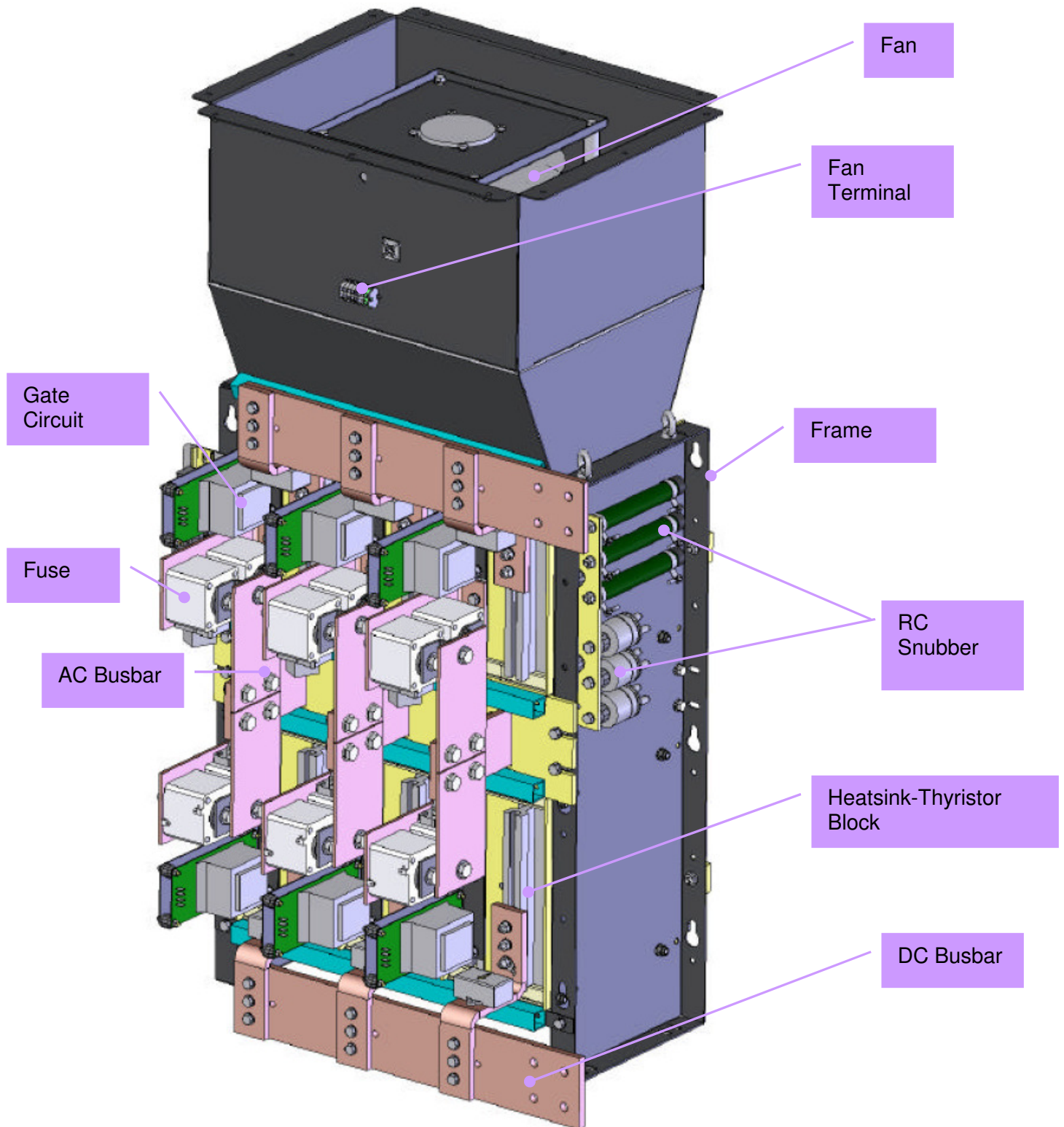
Remove the impeller from the driving shaft end of the old motor. Apply a little grease to the shaft end of the new motor and fit the impeller. Do not forget the key and the spring washer. Small motors are fitted with a split washer.

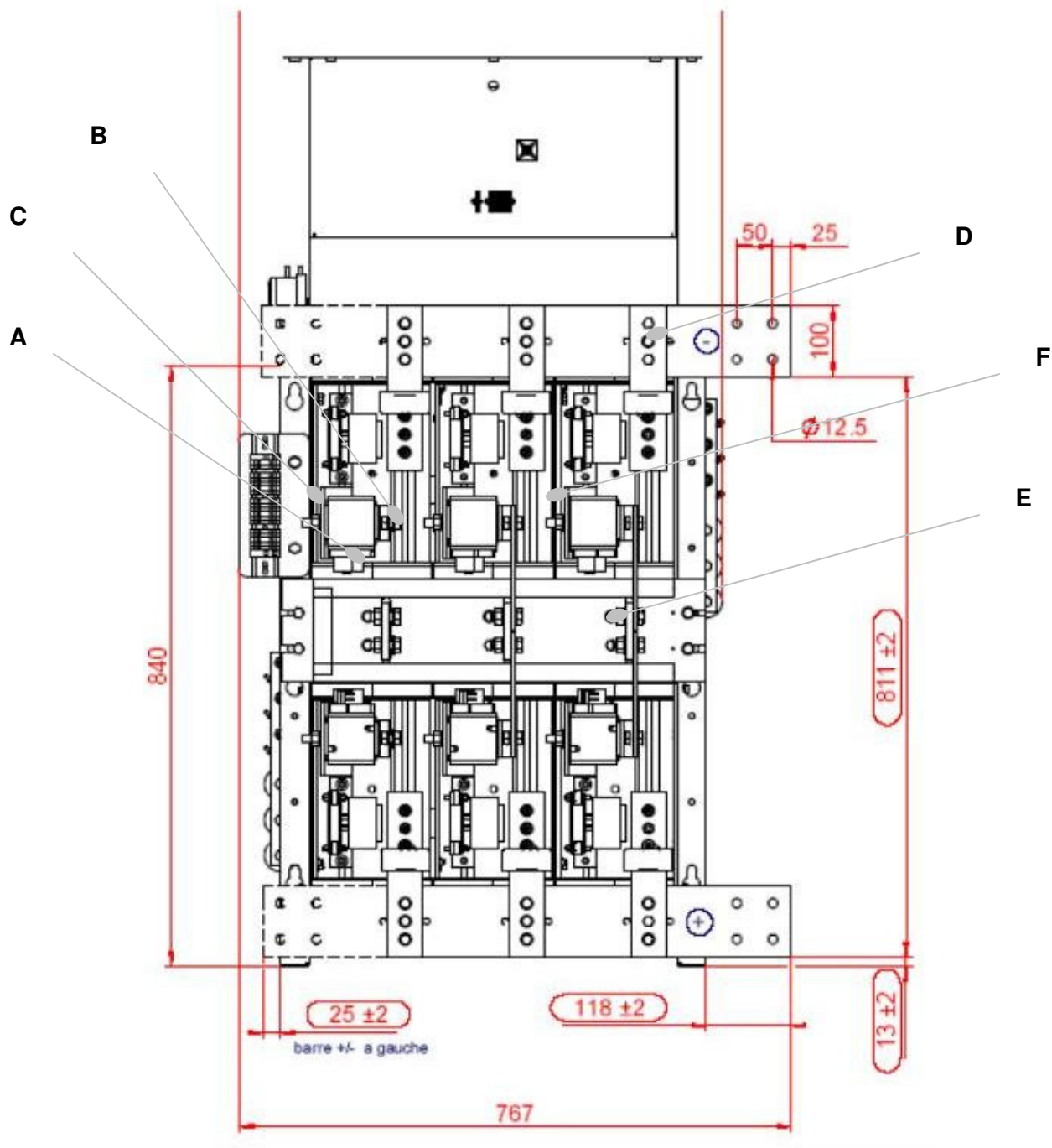
Fit the new motor on the flange, and replace the fan assembly on the cooling duct.

CAUTION: Check that no metallic parts are left in the thyristor bridge, since this may cause substantial damage.



5.6.11 Replacing a block thyristor-heatsink (TYPE S bridge)





Special tools

Thyristor extractor tool, included with each component

Safety requirements

The excitation equipment is switched off and earthed.

Preliminary conditions

No preliminary works are necessary.

Procedure

It is advised to use components of identical VT band for the legs put in parallel
Remove the wires of the RC snubber connected on both heatsinks
Loosen the 3 screws/nuts M8 (D) on the DC busbar
Loosen the 2 screws/nuts M12 (E) on the AC busbar
Now it is possible to extract the block from the frame
Loosen the 3 screws M8 (F) to separate the connection fuse from the heatsink
Before remounting apply thermal contact grease type CG53A (Electrolube) on each fuse face and between connections busbars
Proceed in reverse way for remounting
Respect the tightening torque

5.6.12 Replacing a fuse thyristor (TYPE S bridge)**Special tools**

Wrench

Safety requirements

The excitation equipment is switched off and earthed.

Preliminary conditions

No preliminary works are necessary.

Procedure

It is advised to use fuses with cold resistances similar for the legs put in parallel. The value is mentioned on the fuse case

Take the wires of micro switch off (A)

Loosen the screw/nut on the right side M12 (B)

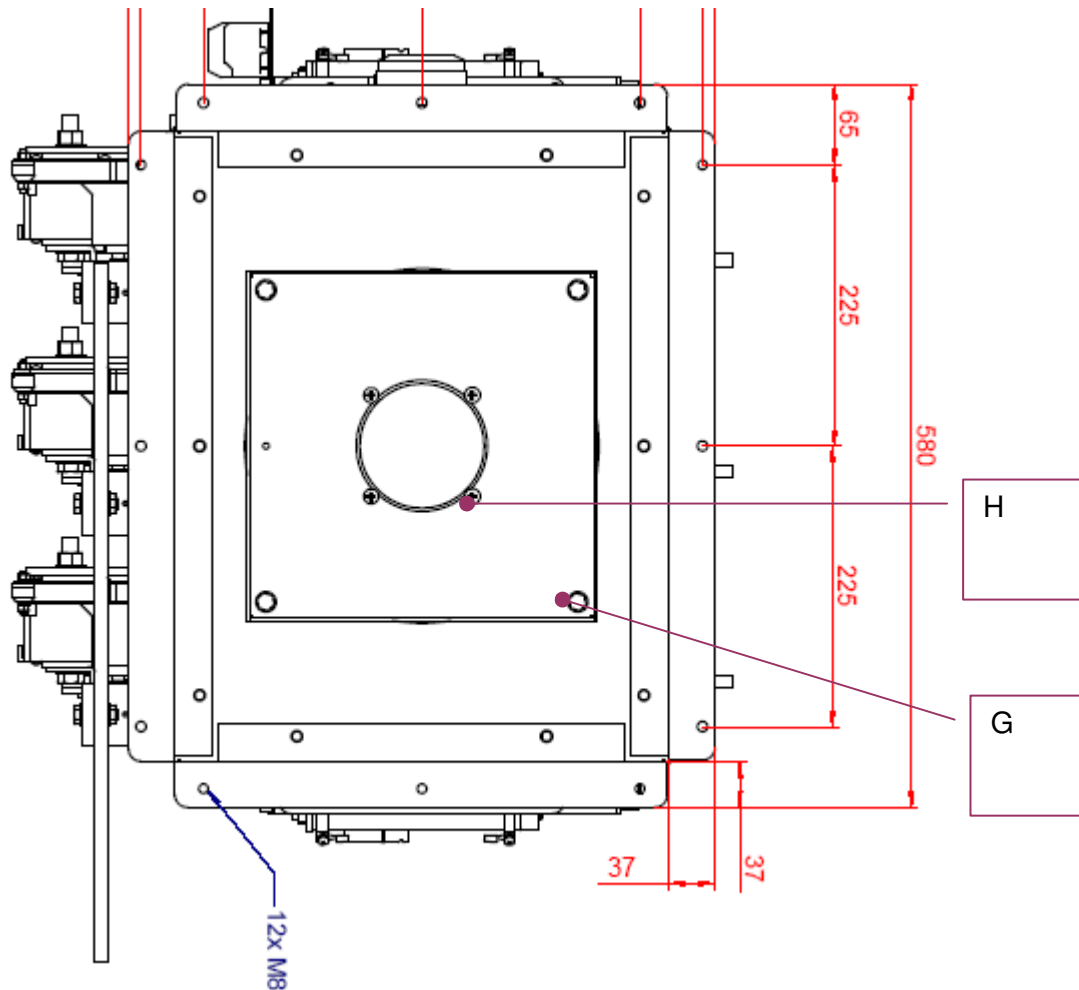
Loosen both nuts M12 on the left side (C) in order to extract the fuse

Before remounting apply thermal contact grease type CG53A (Electrolube) on each fuse face and between connections busbars

Proceed in reverse way for remounting

Respect the tightening torque

5.6.13 Replacing the fan (TYPE S bridge)



Special tools

Wrench

Safety requirements

The excitation equipment is switched off and earthed.

Preliminary conditions

No preliminary works are necessary.

Procedure

Take the wires of the supply connected on X2 off

Loosen the 4 screws M8 (G) at the top side for taking off the fan support.

Now it is possible to extract the fan

Loosen the 4 screws M8 (H) to take the fan away

Proceed in reverse way for remounting